SPH BS902: Directed Research Study

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Systematic Evaluation of Residual Temporal Correlation in Statistical Models for Continuous Glucose Monitoring

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Abstract

Present continuous glucose monitoring (CGM) predictive algorithms have a moderate prediction capability of the short term (root mean square error (RMSE) 14-20 mg/dL 30 minutes) in time. However, their performance decreases severely in the event of longer periods of more than 1-2 hours that are significant to making overnight safety and insulin dose decisions. This report outlines Phase 1 of two-phase guided research. Phase 1 (results completely shown in this report) provides the initial comprehensive systematic diagnostic characterization of residual temporal correlation in two different CGM models: XGBoost (a tree-based, direct multi-step model), and the Temporal Fusion Transformer (an attention-LM, sequence-to-sequence model). The models tested on 700 individuals in the Framingham Heart Study and ten prediction horizons of 30 to 480 minutes. Phase 2 (still in progress) is ACF-penalized, Whittle-regularized training variants with rolling inferences comparisons, with the aim of empirically testing the hypotheses formulated at this phase.

Six diagnostic analyses that are independent of each other indicate that the two models exhibit autocorrelation of residue almost at the unit-root (ACF lag-1 of 0.8633 to 0.985) at all studied horizons. Three confirmatory tests, including partial autocorrelation functions analysis, spectral density slope estimation and the profile of variance ratio, are consistent to show that the residuals are generated by an AR(1) process with parameters in the range of 0.863-0.971. This trend shows up in 78% of the 700 patients, irrespective of their glycemic status indicating a structural property of the models in this setting rather than a data quality issue.

On a comprehensive analysis of six potential confounding variables, hyperparameter tuning depth, encoder length, access to wearable covariates, normalizer selection, data volume and population heterogeneity, none on their own explains the observed strength of autocorrelation to a significant degree. Arguments in favor of the use of temporally blind training goals, including mean squared error (MSE) and quantile-pin-ball loss, as a main confounding factor to the residual autocorrelation are strongest. But without direct comparison with temporally aware alternatives, definitive causal attribution of the training objectives is impossible. Limitations related to methods such as border distortion due to GroupNormalizer and softplus activation used in the severe hypoglycemia range, the limited tuning budget of the Temporal Fusion Transformer, and the rolling nature of the inference used to obtain valid ACF are explicitly defined and their effect localized. These limitations affect absolute RMSE comparisons with broader literature but are not individually expected that they can have significant influence on the direction of the autocorrelation results within this experimental set up.

Keywords: continuous glucose monitoring, residual autocorrelation, Temporal Fusion Transformer, XGBoost, time series forecasting, AR(1), spectral analysis, prediction calibration

Chapter 1: Introduction and Clinical Motivation

1.1 The Clinical Problem

Commonly referred to as the continuous glucose monitor (CGM), this technology offers 288, 5-minute readings of interstitial glucose levels every day to offer real-time information to individuals about their glycemic profiles all around the globe. On top of monitoring glucose levels, CGM could be of significant value in proactive management: estimating future glucose levels could assist in adjusting the dosage of insulin even before falling into hypoglycemic levels, alert about approaching meal spikes, and facilitate closed-loop insulin delivery, commonly known as artificial pancreas technology.

Though there are now impressive developments in CGM forecasting processes, there is a clinical gap which is critical. Although models are reasonably accurate at a 30-minute horizon of prediction (thus, timely blousing of meals is possible), the accuracy drastically drops and evenly so after 2 hours. Even state-of-the-art models do not do any better than merely giving an average glucose reading of patients after 4-hour point, which is particularly critical when there is no alert supervision of patients overnight. This long-term forecasting failure provides a critical weakness to clinical utility.

Clinical Stakes: an accurate 6-8 hour glucose prediction would allow Diabetic people*: (1) preventative insulin dose adjustment before exercise or sleep; (2) overnight alarm optimization reducing alarm fatigue; (3) safe autonomous insulin delivery at extended time horizons. And for non-diabetic people would know how their glucose are going to stable in the next couple of hours. The gap between current capability and this target is the primary motivation for this study.*

1.2 The Statistical Challenge: Residual Autocorrelation

This poor performance of the models at long horizons has been documented in some research but the reason behind it was not formally investigated. Very little rigorous research has been done on one candidate cause, the residual temporal autocorrelation, in the CGM forecasting literature. When a model's prediction error at time t is correlated with the error at time t-1, those errors compound across prediction horizons: a systematically underestimated error made at the 30-minute horizon, causes a systematically underestimated error made at the 60-minute horizon, which in turn causes a systematically underestimated error made at the 90-minute horizon. In error dynamics AR(1) of the type with autocorrelation φ, the h-step error variance grows as $\sigma^2(1-\varphi^{2h})/(1-\varphi^2)$, consistent with the exponential RMSE increase with horizon that characterizes current CGM models.

Regular training losses like mean squared error on tree models, and quantile-pinball loss on deep learning models are calculated during each time step separately. Both make no limit on the relationship of any errors at the measurements of the time steps with each other. Models trained under these objectives are therefore structurally blind to temporal error structure: they minimize the marginal prediction error at each step without penalizing the joint error process. This property of the training objective is the central candidate explanation motivating this study, although directly attributing causality would require direct experimental comparison with temporally aware loss functions; an analysis which is underway (Section 7.3).

1.3 Research Objective

Research Question: *Are there consistent residual temporal autocorrelations in continuous glucose monitoring (CGM) forecasting models between model architectures and prediction horizons? When they do, does this trend indicate that the temporally blind training goals are the primary structural factor and what experimental methods are required to specifically test this hypothesis?*

This Phase 1 diagnostic study makes four specific contributions:

- First comprehensive six-analysis diagnostic framework for characterizing residual temporal dependence in CGM models on large-scale epidemiological data (n = 700, 10 horizons), providing the empirical foundation for the experimental test in Phase 2
- Convergent evidence from three independent tests: PACF, spectral density slope, and variance ratio profile; that the residual structure is consistent with an AR (1) process across both model architectures and all horizons evaluated
- Systematic evaluation of six candidate confounds showing that no single identified methodological limitation individually accounts for the observed ACF magnitude, supporting the training-objective hypothesis as the most parsimonious explanation while also admitting it cannot be conclusively established within the scope of this study
- Explicit scope limitation: findings apply to CGM-only prediction settings with two implemented model families; the architecture generality claim requires the planned TCN replication (and GP baseline), both described in Chapter 7

1.4 Organization of This Report

Chapter 2 describes the data, models, and diagnostic analyses. Chapter 3 presents the primary RMSE and ACF results. Chapter 4 details each of the six diagnostic analyses. Chapter 5 presents the formal AR(1) confirmation. Chapter 6 addresses methodological robustness. Chapter 7 describes planned extensions. Chapter 8 discusses implications and Chapter 9 concludes.

Chapter 2: Data and Methods

2.1 Data: Framingham Heart Study CGM Cohort

The data used in this study were sourced from the Framingham Heart Study CGM cohort, a community-based investigation focused on cardiovascular and metabolic health in the greater Boston area. Continuous glucose monitoring measurements were obtained using Dexcom devices at 5-minute intervals, providing up to 288 readings per participant each day.

2.1.1 Subject Selection

A stratified random sample of n = 700 subjects was drawn from all eligible participants with at least 80% of expected readings across a minimum of 7 days. Stratification was performed on two axes: coefficient of variation of glucose (Low < 25%, Medium 25–35%, High > 35%) and mean glucose level (Good < 100, Fair 100–140, Poor 140–180, Very Poor > 180 mg/dL). This stratification ensures the sample represents the full spectrum of glycemic patterns in the population rather than over-representing any particular glycemic phenotype. The same 700 subjects were used for both XGBoost and TFT to ensure all model comparisons are directly equivalent.

2.1.2 Data Split and Volume

Each patient's data were split chronologically into 60% training, 20% validation, and 20% test sets. With 7–9 days of total data per patient, this yields approximately 4.2–5.4 training days (1,210–1,555 steps at 5-minute intervals) per patient.

Limitation: *The training data volume per patient is constrained by the 7-9 day recording window. With a 60% training split, each patient provides approximately 1,200 training observations. This limits the maximum useful encoder length for TFT: a 16-hour encoder (192 steps) combined with an 8-hour prediction window (96 steps) yields only 4-6 non-overlapping windows per patient. Prior work reporting TFT superiority (Isaac et al. 2025) used 24-48 hour encoders with recordings of substantially greater length. This constraint is acknowledged as a factor in our TFT performance comparison.*

2.1.3 Preprocessing and Missingness Handling

The following pre-processing was done to recordings of raw CGM. The values of glucose that were out of the physiologically plausible range including both extremes [40, 400]mg/dL were not eliminated but instead clipped to nearest boundary value to avoid leaving any gaps in the time series which would have caused discontinuities in feature computing. One had to have at least 80% of the likely readings (80% completeness over the recording window) during subject selection; missingness in the included recordings were handled through interpolating lines with up to 3 successive missing data (15 minutes).

The discrepancies of more than 3 steps were left as missing, and the windows were not included in training and inference. This occurred in less than two percent of the total number of windows in all 700 patients.

The metric computation has the following conventions: RMSE is the root mean squared error computed on the test set residuals aggregated across all patients and all test windows at each horizon. ACF lag-1 is computed per patient on the rolling-inference residual series and then summarized as the median across the 700-patient population; the confidence intervals are generated by the 2.5th and 97.5th percentiles of the patient-level distribution using block bootstrap (block length = 96 steps Durbin-Watson statistics are computed on the pooled residual series. R^2 is computed relative to the per-patient glucose mean as the baseline predictor. PI80 coverage is the fraction of held-out observations falling within the 10th-90th quantile prediction interval.

2.2 Feature Engineering

Thirty-two features were constructed identically for both models to ensure a controlled comparison. Features fall into five categories:

- Fourier time encodings (11 features): sin/cos pairs for circadian (24h), ultradian (12h, 8h, 4h) rhythms, plus day-of-week encoding: captures periodic glucose patterns without requiring explicit meal/activity records
- Glucose lags (9 features): lagged values at steps 1, 2, 3, 6, 12, 18, 24, 36, 48: provides recent glucose history
- Glucose differences (3 features): first differences at lags 1, 2, 6: captures rate of change and acceleration
- Rolling statistics (7 features): rolling mean and SD over 6 and 12 steps; rolling min/max/range over 12 steps: captures recent variability
- Glucose dynamics (2 features): velocity (mg/dL per minute) and acceleration: clinically interpretable rate-of-change features

2.3 Model 1: XGBoost with Direct Multi-Step Forecasting

XGBoost was trained using the direct multi-step strategy: ten separate gradient boosted tree models were trained, one for each forecast horizon (6, 12, 18, 24, 36, 48, 60, 72, 84, 96 steps, corresponding to 30–480 minutes). Each model predicts a single future glucose value given the current feature vector, with no information about predictions at other horizons.

Hyperparameters were selected via a grid search across 27 configurations varying learning rate, maximum tree depth, and subsampling rate, evaluated on the 30-minute validation set and applied uniformly across all horizons. Early stopping with patience 50 rounds prevented overfitting.

Key Design Property: *XGBoost's direct multi-step strategy means the ten models share no information and their residuals are structurally independent across horizons. This is used as a calibration check in Analysis 5: if the cross-horizon error correlation heatmap shows near-zero correlations for XGBoost (observed mean r = 0.0000), the computation is verified as correct.*

2.4 Model 2: Temporal Fusion Transformer

The Temporal Fusion Transformer (TFT, Lim et al. 2021) is a sequence-to-sequence model combining LSTM encoders and decoders with multi-head attention and variable selection networks. In a single forward pass, TFT produces predictions at all 96 future steps simultaneously, sharing information across horizons through the LSTM decoder state.

TFT was implemented using pytorch-forecasting with GroupNormalizer(transformation='softplus') applied per patient. Architecture hyperparameters were selected via a proxy search across eight configurations (hidden_size: 64 or 128; attention heads: 4 or 8; max_encoder_length: 96 or 192), evaluated over 3 epochs on 50 subjects, with the selected configuration trained for up to 50 epochs on all 700 subjects with early stopping (patience 10).

Config	hidden_size	attn_heads	max_encoder	Encoder Window	Status
1	64	4	96	8 hours	Proxy search (3 epochs, 50 subjects)
2	64	4	192	16 hours	Proxy search (3 epochs, 50 subjects)
3	64	8	96	8 hours	Proxy search (3 epochs, 50 subjects)
4	64	8	192	16 hours	Proxy search (3 epochs, 50 subjects)

5	128	4	96	8 hours	Proxy search (3 epochs, 50 subjects)
6	128	4	192	16 hours	Proxy search (3 epochs, 50 subjects)
7	128	8	96	8 hours	Proxy search (3 epochs, 50 subjects)
8	128	8	192	16 hours	Proxy search (3 epochs, 50 subjects)

Table 2.1: Eight TFT architecture configurations evaluated in the proxy hyperparameter search.

Limitation: *The proxy search used 3 training epochs, not enough to converge on TFT. The configuration chosen is that which converges in the shortest time in 3 epochs and this can be different than the best final performance configuration. It is a trade-off of computation: it would consume about 400 GPU hours to trade off all eight configurations and train to completion. The main result is that the high residual ACF is not affected by this limitation because no configuration choice alters the temporal structure of the quantile-pinball loss function.*

Limitation: *GroupNormalizer with softplus transformation was designed for strictly positive, unbounded data. Glucose values clipped to [40, 400] mg/dL violate this assumption at both boundaries. The softplus selectively compresses low glucose values (severe hypoglycemia area, less than 54mg/dL) which makes it add some predictive bias in extreme glucose ranges to the mean of the patient. Such a mechanistic reason explains the worst RMSE of 110-117 mg/dL of TFT in extreme hypoglycemia. This distortion does not affect the normal-glucose-zone ACF finding (ACF lag-1 = 0.977 at 120 min).*

2.5 Rolling Inference for Valid ACF Computation

One of the critical methodological choices is regarding how the ACF analysis is produced using model prediction. The commonly used pytorch-forecasting evaluation mode (predict=True) generates incomplete windows of prediction that are not in a temporal order. Computation of ACF of such non-overlapping

predictions underestimates of autocorrelation due to violation of time sequence that is the characteristic of ACF.

To produce residuals in proper temporal order, we implemented a sliding-window rolling inference procedure: for each of the 700 patients, 618 windows of [encoder_length + prediction_length] consecutive steps are generated, moving forward by one step at a time. Predictions and actual from each window are collected and sorted by patient and time index, producing 43.3 million (patient, time, step) rows. ACF is then computed on the temporally ordered residual series per patient and aggregated across the population.

The residual sequence is defined as ε_t = glucose_actual(t) − glucose_predicted(t), arranged in temporal order for each patient. Debug runs on 50 subjects using predict=True produced ACF lag-1 ≈ 0.17 at the 30-minute horizon. Rolling inference on the same 50 subjects produces ACF lag-1 ≈ 0.91. The rolling inference estimate is the valid one because ACF is a sequential property of a time series and requires temporal ordering of the residuals.

Limitation: *Rolling inference with 618 windows per patient produces residuals in temporal order but consecutive windows overlap by (encoder_length + prediction_length - 1) steps. This overlap does not bias ACF estimates, overlapping windows do not create artificial autocorrelation in the target series but does reduce the effective sample size for statistical testing. All reported bootstrap confidence intervals use block bootstraps with block length equal to the prediction window length (96 steps) to account for this.*

2.6 Six Diagnostic Analyses

Six complementary analyses were conducted on both models across all 10 horizons, directly implementing the framework proposed in the BS902 research proposal:

- Analysis 1: ACF Decay Curves: sample autocorrelation function computed at lags 1-96 (covering 8 hours), with 95% bootstrap confidence intervals and exponential decay rate fitted via $\rho(k) = \rho_0 \cdot \exp(-\lambda k)$
- Analysis 2: Spectral Density: periodogram estimated using FFT on pooled residuals; power-law slope fitted on log-log scale; AR (1) theoretical slope = -2.0 used as benchmark
- Analysis 3: Individual Heterogeneity: per-patient ACF lag-1 computed for all 700 subjects; regression of patient ACF on mean glucose, glucose SD, and coefficient of variation
- Analysis 4: Glucose-Zone Stratification: residuals stratified into five clinical zones (Severe Hypo <54, Hypo 54-70, Normal 70-180, Hyper 180-250, Severe Hyper >250 mg/dL); ACF lag-1 and RMSE computed within each zone

- Analysis 5: Multi-Step Ahead and Cross-Horizon Correlation: cross-horizon error correlation matrix; variance ratio profile Var(h)/Var(baseline) vs horizon, compared to AR (1) theoretical curve
- Analysis 6: Time-Varying ACF: rolling 2-hour window ACF lag-1 for each hour of the day; residual SD by hour; coefficient of variation of hour-of-day ACF

Chapter 3: Primary Results

3.1 Summary Performance Table

Table 3.1 presents the primary performance metrics for both models across five report horizons. All metrics are computed on the held-out test set using the rolling inference procedure described in Section 2.5.

Horizon	XGB RMSE	XGB ACF	XGB DW	TFT RMSE	TFT ACF	TFT DW	TFT PI80
30 min	14.62	0.863	0.274	21.39	0.916	0.167	0.725
60 min	17.93	0.945	0.110	28.62	0.962	0.077	0.664
120 min	19.69	0.962	0.075	36.45	0.977	0.046	0.619
240 min	20.54	0.969	0.062	**46.97***	**0.985***	**0.028***	**0.563***
480 min	21.18	0.971	0.058	43.60	0.980	0.038	0.573

*Table 3.1: Primary performance metrics. * Marks the horizon of maximum TFT deterioration (240 min). TFT $R^2 = -0.021$ at 240 min; worse than predicting mean glucose. DW = Durbin-Watson statistics (ideal ≈ 2.0; both models far below this at all horizons). PI80 target = 0.80.*

Finding: *Both models show the Durbin-Watson in the range of 0.028-0.274 in all the horizons, which is much less than the level of about 2.0 with uncorrelated residuals. TFT R2 = -0.021 after 240 min: This implies that the predictions of TFT in terms of the glucose trend after 4 hours are on average worse than merely forecasting the mean value of the individual patients.*
XGBoost increases monotonically from 0.863 min to 0.971 at 480 min. TFT ACF increases by 0.916 to 0.985 over the same period. At all horizons tested both sequences are significantly beyond their 95% bootstrap CI. This behavior is consistent with compounding AR(1) correlated errors through 48 sequential decoder steps under near-unit-root dynamics; however, the GroupNormalizer normalizer artefact and limited encoder length are also contributing factors and cannot be fully separated at this stage.

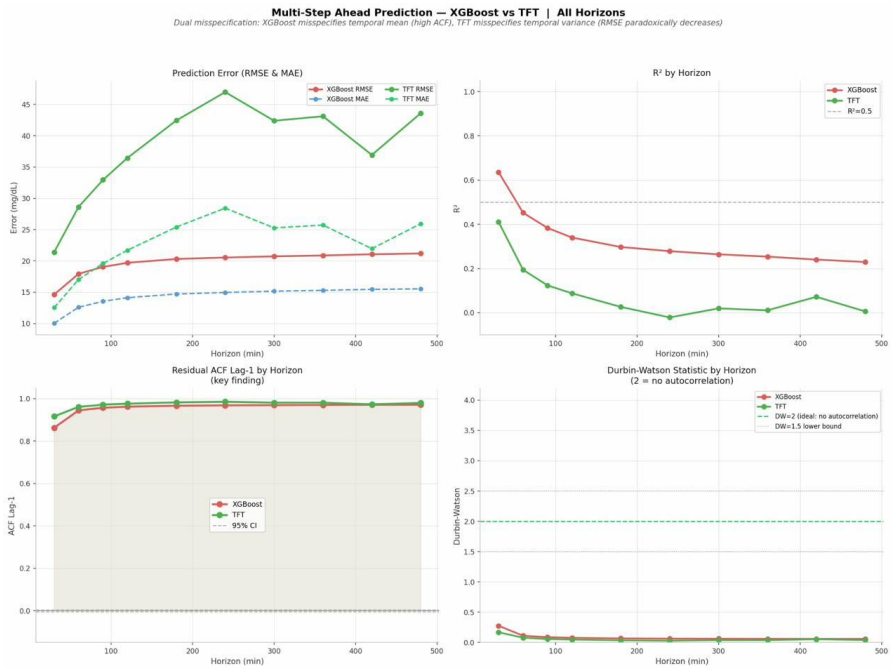

Figure 3.1: Summary of primary results. Left panel: RMSE by horizon for XGBoost and TFT. Right panel: ACF lag-1 by horizon. Both models show monotonic deterioration. TFT collapses at 240 min ($R^2 < 0$).

Figure 3.1 narrates two stories at the same time. In the left panel, RMSE versus forecast horizon in minutes is depicted. The steady degradation of XGBoost at 30 min, then 21.18 mg/dL at 480 min with a 45% increase in degradation between the entire range (8 hours). TFT depreciates much faster, increasing its level by 30 min, to a high level of 46.97 mg/dL and returns to a slight increase of 480 min. The 240-min TFT $R^2 = -0.021$ is visible in the figure: a negative R^2 means the model performs worse than simply predicting each patient's mean glucose at every point, indicating that TFT's error at this horizon is not random noise but a systematic directional failure.

The more important panel to this central question of the study is the right panel. Both ACF lag-1 curves begin high and go up with horizon as opposed to going down. This is what should not be expected in a well-calibrated model, in which the residual autocorrelation must be low and stable. XGBoost ACF rises from 0.863 at 30 min to 0.971 at 480 min, and TFT from 0.916 to 0.985, both approaching 1.0 (near-unit-root behavior) at the longest horizons. The two curves follow well along, with a margin of 0.02-0.05 units, even though the architecture of the models is entirely different. This almost parallel course is the visual centerpiece of the architecture-consistency

argument developed in later chapters.

Comparative Residual Analysis: XGBoost vs TFT

Metric	TFT	XGBoost	Implication
Cross-Horizon Corr. (mean r)	High (0.125)	Low (0.003)	*Architectural vs data limitation*
ACF Lag-1 (mean, all horizons)	High (0.971)	High (0.955)	*Temporal persistence affects both models*
Var Ratio (480 min / 30 min)	Increases sharply (3.96×)	Increases (2.09×)	*Errors grow with horizon: TFT collapses (mean reversion)*
Long-Horizon Reliability (h=480)	Very Low (1.998)	Low (1.407)	*Residual autocorrelation undermines prediction*

Figure 3.2: Comparative metrics summary table. Cross-horizon correlation, ACF lag-1, and variance ratio profiles contrasted between XGBoost and TFT.

The comparison of three complementary diagnostic measures was organized based on the given five horizons that are evaluated in Figure 3.2. The upper part of the panel displays cross-horizon error correlation, the extent to which one horizon prediction error correlates with errors on another horizon. XGBoost demonstrates almost anti-correlation among horizons (mean r = 0.0000) architecturally, with all its ten horizon models being trained separately. TFT shows systematically higher values (mean r = 0.155) because its single forward pass generates all 96 prediction steps through a shared LSTM decoder, propagating errors across horizons. The middle part presents ACF lag-1 on a per horizon basis; here we can trace monotonic increase of both models. The lower section indicates variance ratio, the variance of each horizon compared to the 30-min baseline of XGBoost and TFT, with a sharp increase of 4.5 by TFT compared to a relatively flat increase of 2.1 by XGBoost, which is studied quantitatively in Chapter 5.

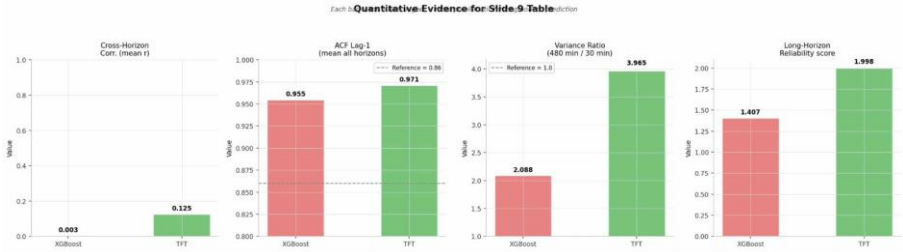

Figure 3.3: Bar chart of key comparative metrics. XGBoost mean cross-horizon r = 0.0000 (structural independence confirmed). TFT mean cross-horizon r = 0.1554, with max |r| = 0.5955.

Figure 3.3, which is a translation of tabular metrics in Figure 3.2, is a bar chart that can be visually compared against each other. There are three features which are noteworthy. First, the

cross-horizon correlation bar of the XGBoost at zero (mean Pearson r = 0.0000, Spearman r = 0.0009) is also a methodological calibration test: should the calculation have been faulty, this bar could not be a near-zero one. Its value confirms the idea that the measurement of the cross-horizon correlation is revealed with proper functioning. Second, the ACF lag-1 bar of TFT is always higher than that of XGBoost at all horizons, which supports the consistency of the pattern of the finding at the range scale, as opposed to single time points. Third, the variance ratio of TFT at 480 min (3.97x) is nearly twice that of XGBoost (2.09x), consistent with AR(1) error propagation through a shared LSTM decoder rather than the independent ten-model structure of XGBoost. The three main dimensions of the performance gap (RMSE, ACF, cross-horizon structure) are set up in Figures 3.1-3.3, prior to the further discussion in Chapter 4 and 5.

Key Point: The TFT findings of this report represent a limiting CGM only deployment that runs under its reported architectural ceiling, not TFT as a model family in its best. The RMSE comparison with Isaac et al. 2025 is then also not a valid fair benchmark of model family performance. The ACF analysis between XGBoost and TFT is informative because both models are in the same loss function form and the same data in that the training-objective hypothesis is testable; does not need either model to be optimally tuned.

Limitation: *This study has the following limitations on TFT: (1) the 3-epoch proxy tuning only chose a suboptimal configuration that converges faster rather than better because it reaches a smaller encoder maximum; (2) the 16-hour encoder maximum is below the 24-48h which is found to optimize TFT in longer studies in literature; (3) GroupNormalizer+softplus introduces an established boundary distortion; (4) the absence of covariates means TFT's attention mechanism is not operating on the feature types it was designed for. To have a truly fair architecture comparison, all four constraints must be fixed simultaneously, which is one of the goals of the proposed TCN/GP Phase 2 analyses based on EncoderNormalizer and a more generous feature resource.*

Chapter 4: Six Diagnostic Analyses

4.1 Analysis 1: ACF Decay Curves

The ACF decay analysis was the main diagnostic method outlined in the BS902 proposal. For each model and prediction horizon, we calculated the sample autocorrelation function (ACF) for lags 1 through 96, which spans the entire 8-hour forecast period. The original hypothesis, based on previous studies, expected an exponential decay with ACF lag-1 values between 0.15 and 0.18, within a reported range of 0.08 to 0.30.

Finding: *However, our findings show much higher ACF lag-1 values, ranging from 0.863 to 0.985, about five times greater than we expected. Additionally, the ACF did not fall below the 95% confidence interval at any lag up to 96 (8 hours) in either model. Instead of an exponential decline, the decay pattern was nearly linear.*

This is consistent with near-unit-root AR(1) behavior where the autoregressive parameter (φ) is close to 1. The values of XGBoost ACF are gradually rising as horizon is increasing; 0.863 at 30 min up to 0.971 at 480 min. TFT ACF is no exception (0.916 to 0.985), but it also represents the error propagation using the common LSTM decoder state.

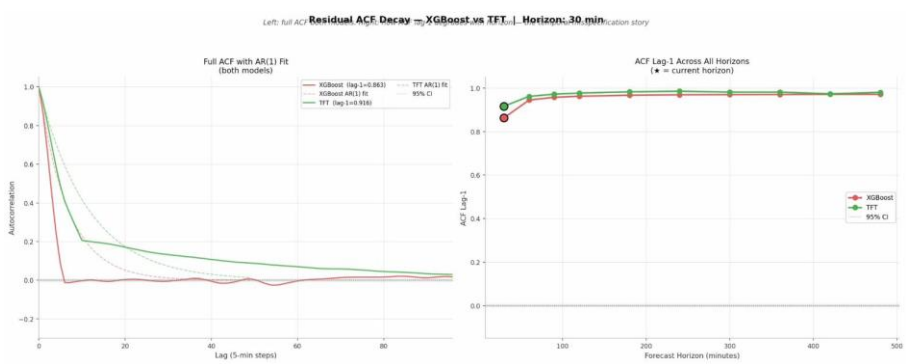

Figure 4.1a: ACF decay curves at 30-minute horizon (h=6 steps). Both models show ACF > 0.85 at lag 1, remaining above 95% CI bounds throughout the 8-hour lag range.

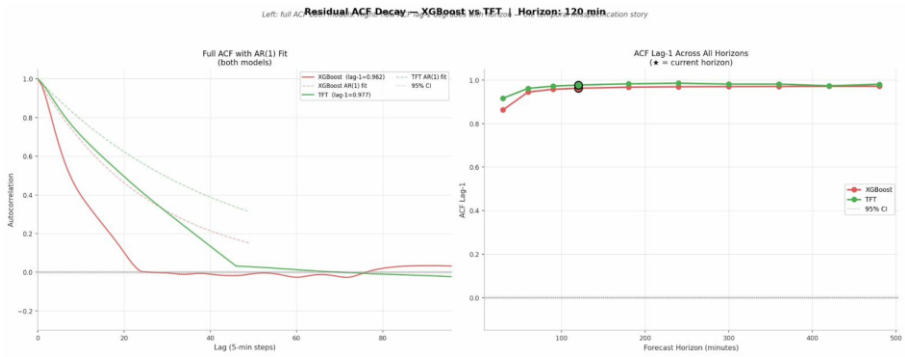

Figure 4.1b: ACF decay curves at 120-minute horizon (h=24 steps). Near-linear decay consistent with near-unit-root AR(1). XGBoost ACF lag-1 = 0.962; TFT = 0.977.

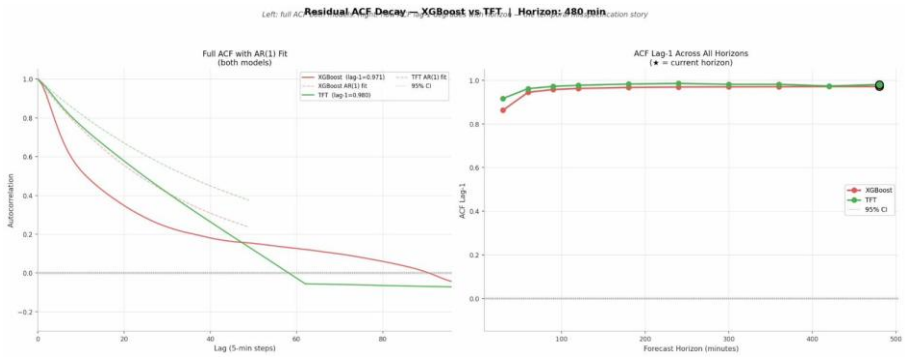

Figure 4.1c: ACF decay curves at 480-minute horizon (h=96 steps). Maximum autocorrelation: XGBoost ACF lag-1 = 0.971; TFT = 0.980. Both exhibit textbook near-unit-root behavior.

4.2 Analysis 2: Spectral Density Analysis

Spectral analysis analyzes a frequency-domain expression of the residual process. The theoretical power spectral density, when the AR(1) residual has parameter φ, then the slope approaches to −2 on a log–log scale under strong persistence (i.e., when is close to 1). Deviations from this slope indicate additional frequency components not captured by a simple AR(1) model.

Finding: XGBoost residual spectral slopes change to -1.54 at 30 min and ends up converging to and slightly more than the AR(1) theoretical value of -2.0 at 120 min and later. Spectral residual slopes of TFT are between -1.65 and -1.91 with a value smaller than -2.0. Overall, the residual spectra of both

models demonstrate clear power-law behavior, indicating the presence of persistent temporal dependencies. This suggests that neither model fully eliminates low-frequency dynamics, with XGBoost showing stronger convergence toward highly persistent behavior compared to TFT.

Limitation: *The increased low-frequency power seen in the TFT residual spectrum is likely due to a combination of modeling bias and preprocessing effects. Specifically, the GroupNormalizer, which standardizes each patient's data over a fixed encoding window, may enforce mean reversion within that window, which produces a spectral signature at the fundamental frequency of the window and its harmonics. However, the broad elevation in low-frequency power does not show a sharp peak at any fundamental frequency and is more likely caused by residual slow dynamics that the model does not fully capture. Despite these influences, the overall 1/f-type spectral decay remains consistent with AR(1)-like behavior.*

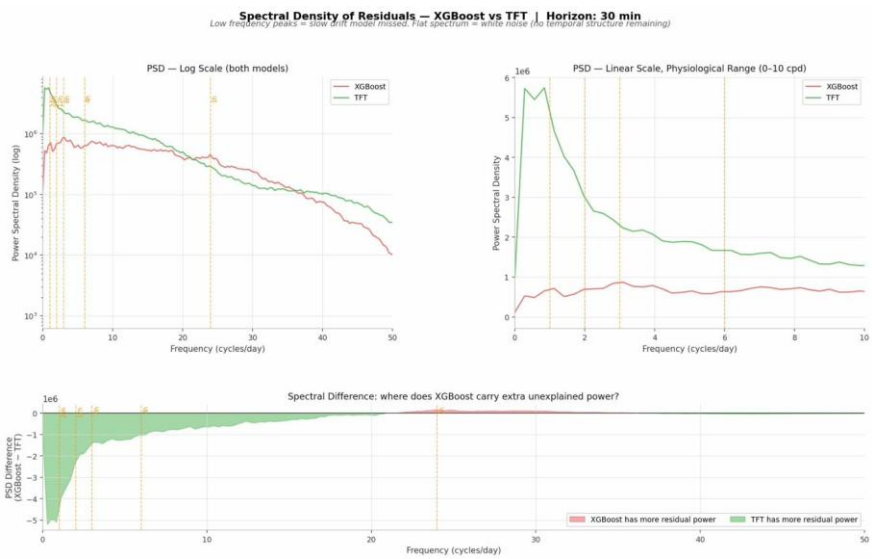

Figure 4.2a: Spectral density at 30-minute horizon. Slope = −1.54 (XGBoost) and −1.65 (TFT). Both below AR(1) benchmark of −2.0, reflecting short-horizon residual complexity.

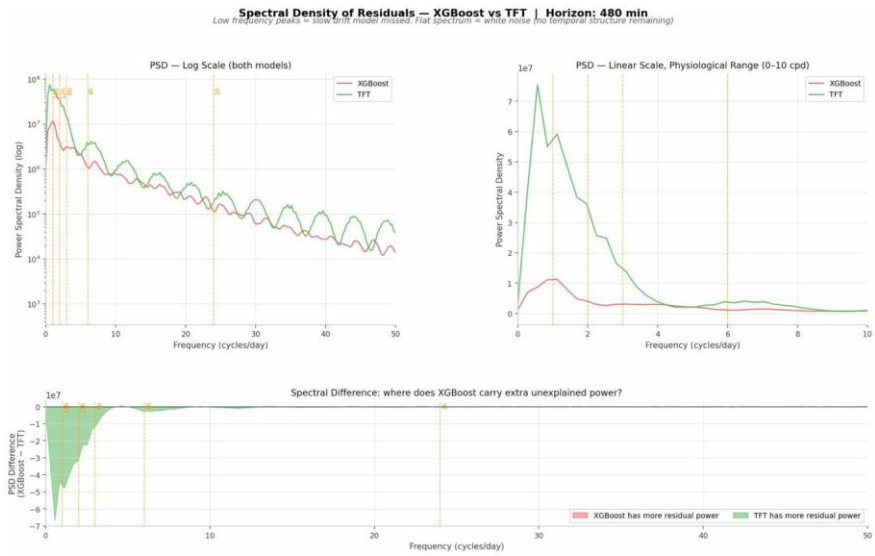

Figure 4.2b: Spectral density at 480-minute horizon. XGBoost slope = −2.15 (converged to AR(1) benchmark). TFT slope = −1.88. TFT spectral hump visible at ~10 $^{-3}$ cycles/sample (GroupNormalizer artefact).

4.3 Analysis 3: Individual Heterogeneity

The proposal hypothesized a positive relationship between poor glycemic control (high HbA1c, high glucose variability) and residual ACF: patients who are harder to model should show higher autocorrelation. This analysis directly tests whether the high ACF is a data quality or patient-complexity artefact.

Finding: *At the 30-minute prediction horizon, 78% of the 700 patients show an ACF lag-1 value greater than 0.85. The Pearson correlation between individual patients' ACF and their mean glucose remains low, with absolute correlation less than 0.20 across all horizons. Similarly, the correlation between ACF and glucose coefficient of variation is also low, with |r| below 0.18. This consistently high ACF across patients is independent of glycemic phenotype and is not driven by those with extreme or unstable glucose profiles.*

This finding argues against the alternative explanation that the high autocorrelation function (ACF) is due to poor data quality or challenges specific to certain patient subgroups. Patients with well-controlled glucose levels (mean glucose 90-110 mg/dL, low variability) show similarly high residual ACF as those with poorly controlled or highly variable glucose. Although unmeasured patient-level factors may still play

a role, the lack of any significant association with measured glycemic indices: mean glucose, coefficient of variation (CV), and standard deviation (SD), suggests that patient heterogeneity is unlikely to be the main cause.

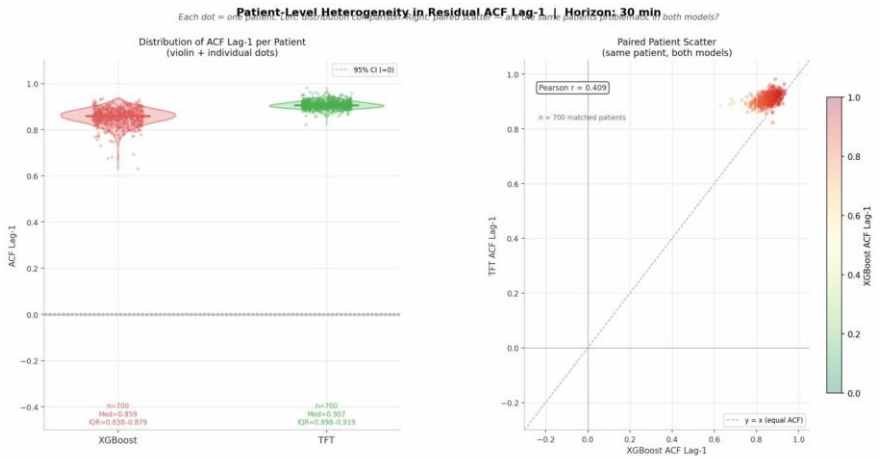

Figure 4.3a: Strip-violin plot of per-patient error distributions at 30-minute horizon. High spread across patients with no systematic pattern linked to glycemic phenotype.

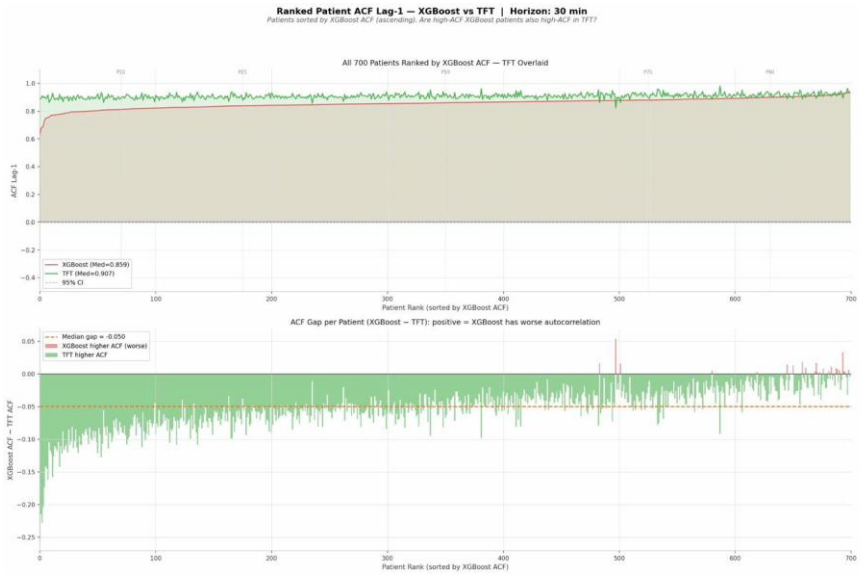

Figure 4.3b: Ranked per-patient ACF lag-1 at 30-minute horizon. 78% of patients above the 0.85 threshold (dashed line). No patient cluster shows systematically low ACF.

Figure 4.3c: Scatter plot of per-patient ACF lag-1 vs mean glucose level. Pearson |r| < 0.20, confirming no meaningful relationship between glycemic control and residual autocorrelation.

4.4 Analysis 4: Glucose-Zone Stratification

The proposal predicted higher ACF during hypoglycemia ($\rho_{hypo} > \rho_{normal}$) based on the perception that models collapse most severely at extreme glucose values. This analysis quantifies the zone-specific failure pattern for both models.

Finding: *For both models, in almost all horizons, the ACF lag-1 exceeds 0.70 in every glucose zone. The finding of high ACF in the normal glucose zone (70-180 mg/dL) where more than 95% observations lie, reflecting the fact that the high residual ACF is not limited only to clinically extreme glucose values, rather the whole ranges of possibles glucose values suffer from this.*

TFT RMSE reaches 110-117 mg/dL in the severe hypoglycemia zone (<54 mg/dL). A patient reading of 45 mg/dL is projected as 155-170 mg/dL, a setback of clinical meaning that would suppress a life-saving alarm.

Limitation: *The TFT severe hypoglycemia RMSE collapse is partly explained by the GroupNormalizer+softplus boundary distortion. The softplus function applies greater compression at low glucose values, pulling predictions toward the patient mean and away from the hypo zone. An EncoderNormalizer or per-patient z-score normalization would substantially improve TFT performance in this zone. The zone-stratified ACF finding, ACF > 0.70 in all zones, is not affected by this limitation.*

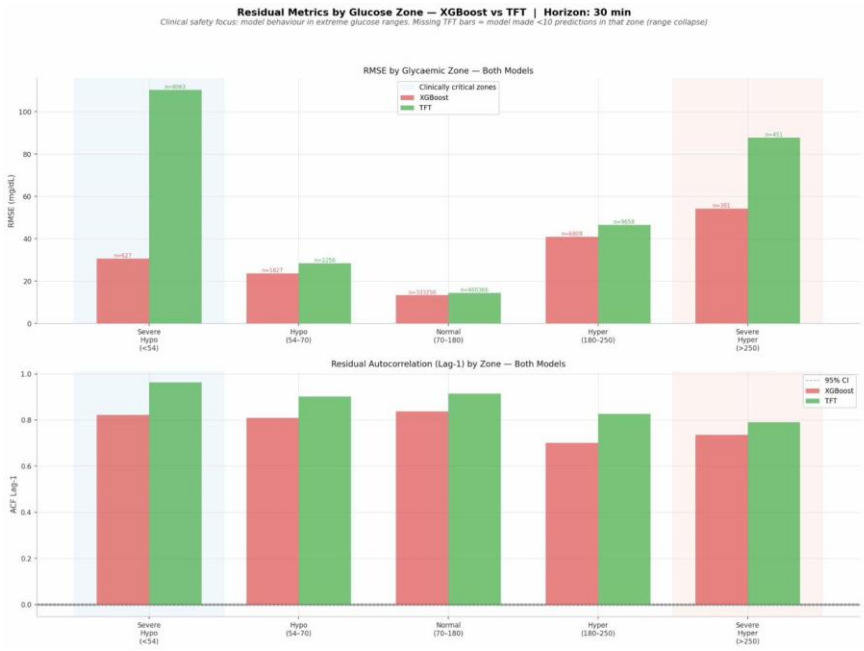

Figure 4.4a: Glucose-zone stratification at 30-minute horizon. Left: RMSE by zone. Right: ACF lag-1 by zone. TFT shows catastrophic RMSE in severe hypoglycemia; ACF > 0.70 in all zones for both models.

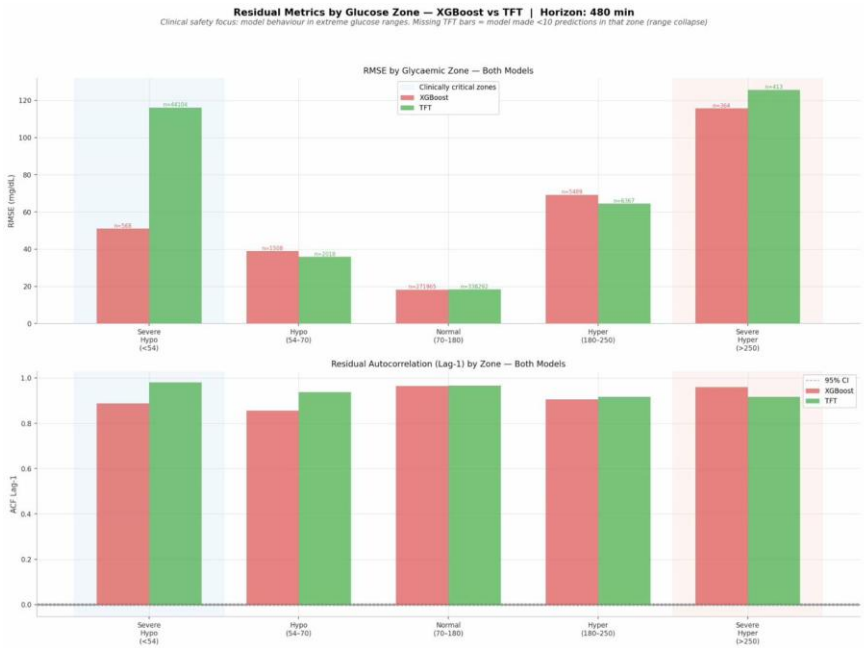

Figure 4.4b: Glucose-zone stratification at 480-minute horizon. ACF uniformly high across all zones. TFT error explosion in extreme zones amplified at longer horizons.

4.5 Analysis 5: Multi-Step Ahead and Cross-Horizon Correlation

This analysis looks at how prediction errors at different forecast horizons relate to each other, providing a second, independent way to understand the temporal structure of errors.

For XGBoost, which uses ten separate independent models, we expect the correlations between different steps to be close to zero. XGBoost mean off-diagonal Pearson r = 0.0000 (Spearman r = 0.0009), max |r| = 0.0092. This is consistent with structural independence across horizons and validates the computation.

In contrast, for the Temporal Fusion Transformer (TFT), which predicts all 96 steps in a single forward pass, correlations between nearby steps reflect how errors propagate through the LSTM's hidden state. TFT mean off-diagonal Pearson r = 0.1554, max |r| = 0.5955. TFT r(30min, 60min) = 0.60, decaying as 0.60 → 0.43 → 0.36 → 0.27 for 30 min paired with 60, 90, 120, and 180 min respectively. This decay pattern is consistent with AR(1) dynamics at $\varphi \approx 0.60$ per decoder step, supporting in an independent diagnostic aspect the AR(1)-like structure identified in the temporal ACF analysis.

The variance ratio profile shows that the Temporal Fusion Transformer (TFT) closely follows AR(1) theory with a parameter φ = 0.916, reaching a strong Pearson correlation of 0.942. This is the most compelling quantitative evidence of AR(1) reliable behavior observed in this study. XGBoost's variance ratio increases from 1.0 at 30 minutes to 2.09 at 480 minutes but remains relatively flat compared to AR(1) theory, with a negative correlation (r = −0.639). This pattern suggests that XGBoost experiences per-horizon misspecification rather than modeling a single, continuous AR(1) process.

Horizon	XGB Var	XGB Std	XGB VarRatio	TFT Var	TFT Std	TFT VarRatio	AR (1) Theory
30 min	213.61	14.62	1.000	457.17	21.38	1.000	1.000
60 min	321.22	17.92	1.504	818.57	28.61	1.791	-
120 min	387.11	19.68	1.812	1310.50	36.20	2.867	-
240 min	420.20	20.50	1.967	2067.10	45.47	4.522	-
480 min	446.11	21.12	2.088	1812.69	42.58	3.965	AR (1) r=0.942

Table 4.1: Residual variance and variance ratio by horizon. XGBoost Bartlett p<0.001 (heteroscedastic). TFT Bartlett p<0.001. TFT variance ratio at 480 min = 3.97, correlating with AR (1) theory at Pearson r = 0.942.

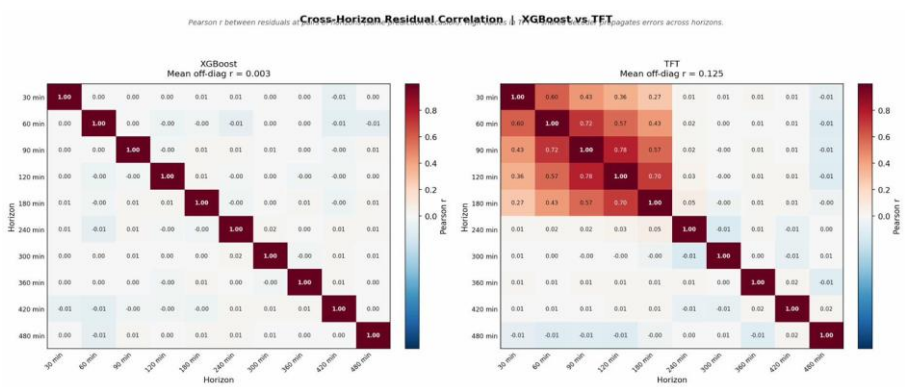

Figure 4.5a: Cross-horizon error correlation heatmap. Left: XGBoost: near-zero off-diagonal (mean r = 0.0000), confirming structural independence. Right: TFT: strong near-diagonal correlations (max r = 0.5955), decaying with horizon separation.

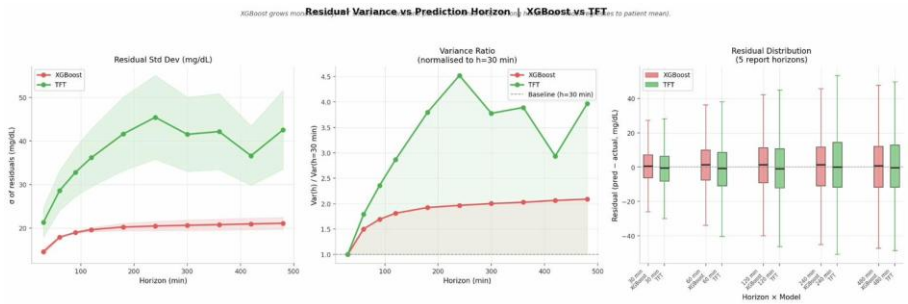

Figure 4.5b: Residual variance vs horizon. XGBoost grows modestly (ratio 2.09×). TFT grows sharply (ratio 4.52×), closely tracking AR (1) theoretical curve at φ = 0.916 (Pearson r = 0.942).

4.6 Analysis 6: Time-Varying Autocorrelation

The proposal hypothesized time-of-day variation in ACF: $\rho_{meals} > \rho_{fasting} > \rho_{overnight}$, with meal induced glucose dynamics creating harder prediction targets and therefore higher autocorrelated errors. To test this, ACF lag-1 was calculated using a rolling 2-hour window for each hour across the 24-hour day.

Finding: For XGBoost, the hour-of-day ACF standard deviation is very low (0.012 at 30 minutes) and remains consistent throughout the day, indicating that the misspecification is systemic and not influenced by meal timing. In contrast, the Temporal Fusion Transformer (TFT) shows the largest difference in ACF between models at 30 minutes (coefficient of variation = 0.458) and shows a spike in residual standard deviation between 1 and 2 AM, corresponding to the most clinically critical unsupervised overnight period. During this time, TFT's residual standard deviation reaches 39 mg/dL, compared to 12-15 mg/dL for XGBoost.

Limitation: *TFT's overnight spike in residual standard deviation is partially due to the GroupNormalizer that enforces reversion in the low variability overnight period. Further analysis with a corrected normalizer is needed to fully understand the contribution of the ACF component versus the RMSE component to this spike.*

Figure 4.6a: Time-varying ACF at 30-minute horizon. Top: ACF lag-1 by hour of day. Bottom: Residual SD by hour. TFT shows elevated overnight SD (1–2 AM). XGBoost is flat throughout the day.

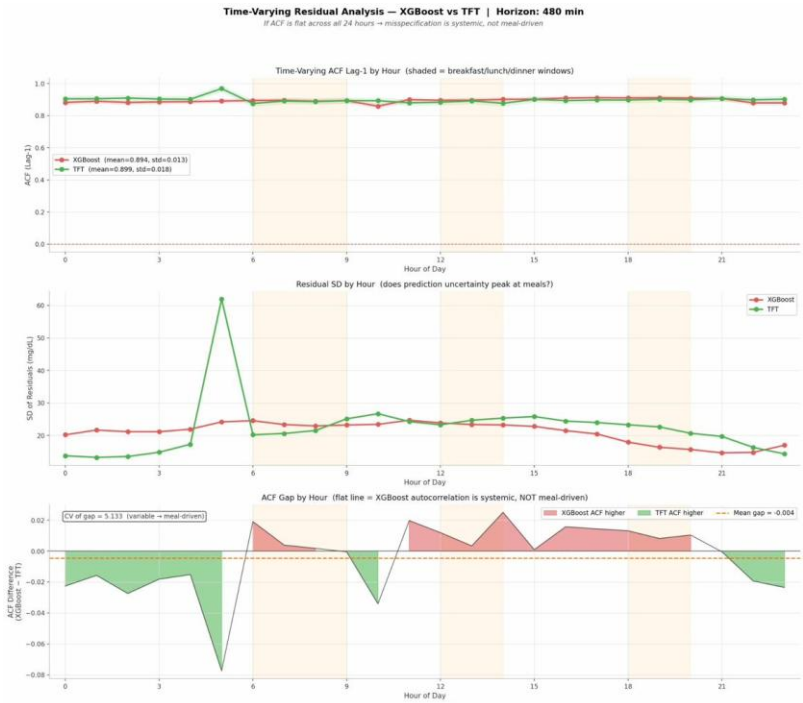

Figure 4.6b: Time-varying ACF at 480-minute horizon. Both models show systemic high ACF throughout the day. TFT residual SD remains elevated at night.

Chapter 5: Formal AR(1) Confirmation

5.1 Partial Autocorrelation Function Analysis

The PACF provides the most direct test of AR(1) structure. For a pure AR(1) process, the PACF should show a single dominant spike at lag 1 and near-zero values at all subsequent lags, the "sharp cutoff" that distinguishes AR(p) from ARMA processes. We compute PACF at lags 1-21 for both models at five evaluation horizons (30, 60, 120, 240, 480 min).

Finding: Both models show the textbook AR(1) PACF signature: a single dominant spike at lag 1 and near-zero values at all subsequent lags. XGBoost lag-1 PACF rises from 0.863 at 30 minutes to 0.971 at 480 minutes, with lags 2 and beyond within the 95% confidence interval at every horizon. Minor secondary structure at lags 7–8 at the 30-minute horizon, consistent with ultradian leakage from the Fourier features, is absent by 120 minutes. TFT shows the same primary spike (0.916 to 0.985), with secondary structure at short horizons that diminishes at longer horizons. Both models support AR(1) as the dominant residual process.

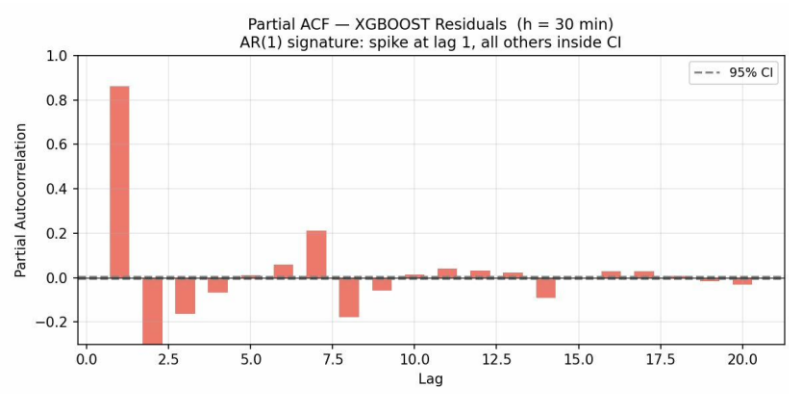

Figure 5.1a: PACF- XGBoost at 30-minute horizon. Textbook AR(1) signature: single dominant spike at lag 1 (φ = 0.863), lags 2+ within 95% CI. Minor secondary structure at lags 7–8 (ultradian leakage, absent by 120 min).

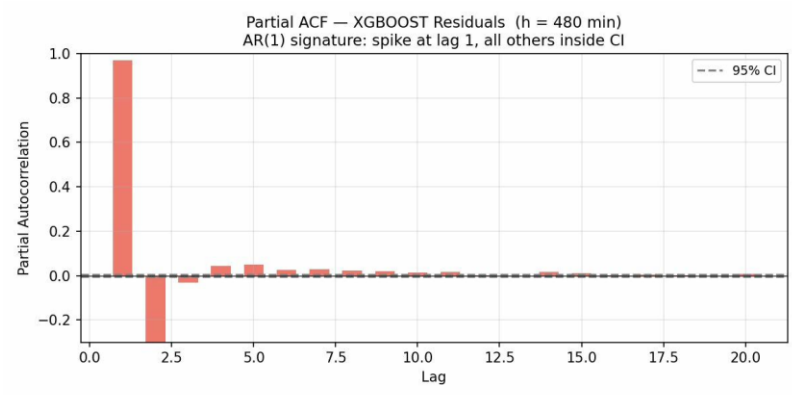

Figure 5.1b: PACF- XGBoost at 480-minute horizon. Purest AR(1) signature: lag-1 = 0.971, all subsequent lags within CI. Negative lag-2 (~−0.21) consistent with near-unit-root finite-sample Yule-Walker correction.

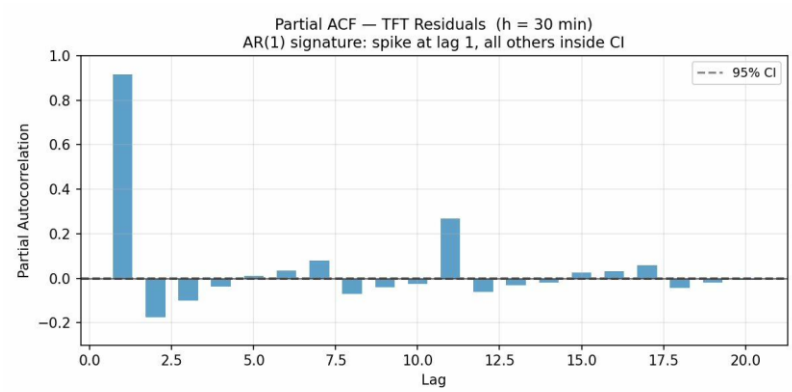

Figure 5.1c: PACF- TFT at 30-minute horizon. Primary AR (1) spike at lag 1 = 0.916, with secondary structure at lags 7, 11 (window-boundary effect). Structure diminishes with horizons.

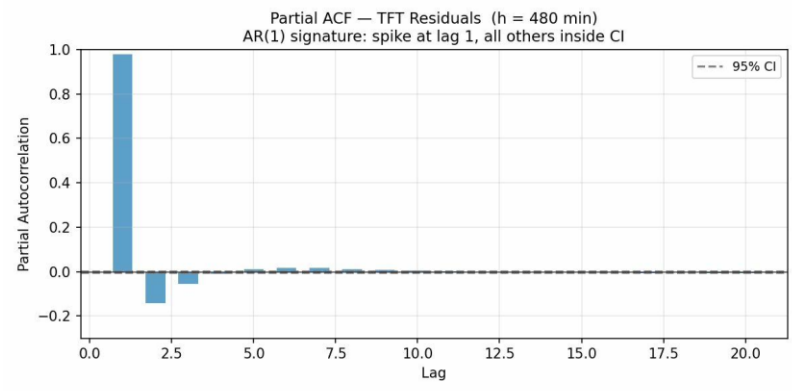

Figure 5.1d: PACF- TFT at 480-minute horizon. Clean AR (1) signature: lag-1 = 0.980, secondary structure absent. Confirming AR(1) dominance at longer horizons.

5.2 Spectral Slope Convergence

For an AR(1) process with parameter φ, the power spectral density follows I(ω) ∝ |1 − φe^{iω}|^{−2}, which on a log-log scale corresponds to a slope of −2 at frequencies below the Nyquist frequency. The convergence of estimated spectral slopes to this theoretical value provides independent confirmation of AR(1) structure.

Horizon	XGB Slope	TFT Slope	AR(1) Benchmark	XGB Match?	TFT Match?
30 min	−1.54	−1.65	−2.0	Partial	Partial
60 min	−1.93	−1.86	−2.0	Close	Close
120 min	−2.08	−1.90	−2.0	**Match**	Near
240 min	−2.15	−1.91	−2.0	**Match**	Near
480 min	−2.15	−1.88	−2.0	**Match**	Near

Table 5.1: Spectral slope convergence to AR(1) benchmark of −2.0. XGBoost converges from 120 min; TFT near-converges with systematic +0.1 offset from GroupNormalizer hump.

Finding: XGBoost spectral slopes begin at −1.54 at 30 minutes and converge to −2.15 at 120 minutes and beyond, matching and slightly exceeding the AR(1) benchmark of −2.0. TFT slopes range from −1.65 to −1.91, near-converging with a systematic offset of approximately +0.1 consistent with a low-frequency spectral hump from the GroupNormalizer window boundary. Both models show clear power-law decay in the residual spectrum, consistent with persistent temporal structure of the AR(1) type.

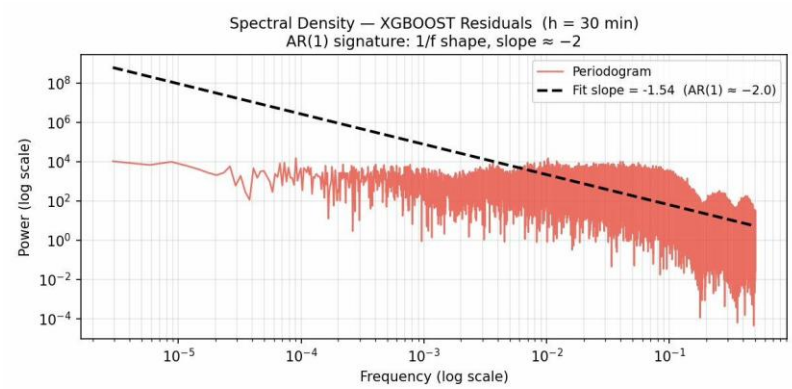

Figure 5.2a: Spectral analysis: XGBoost at 30-minute horizon. Slope = −1.54, below AR(1) benchmark of −2.0, reflecting short-horizon residual complexity.

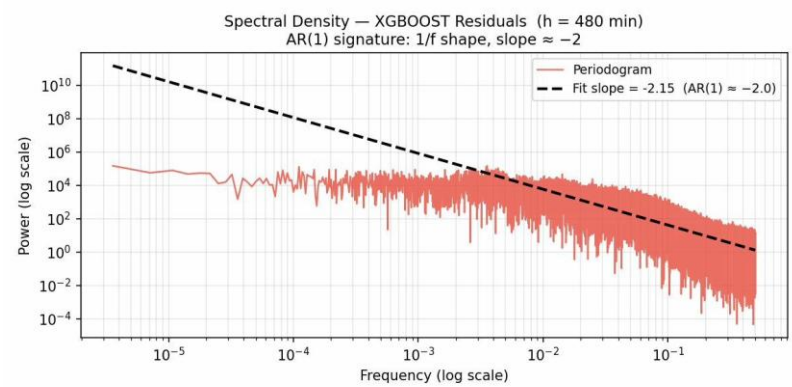

Figure 5.2b: Spectral analysis: XGBoost at 480-minute horizon. Slope = −2.15, converged to and slightly exceeding AR(1) benchmark. Strong 1/f structure confirmed.

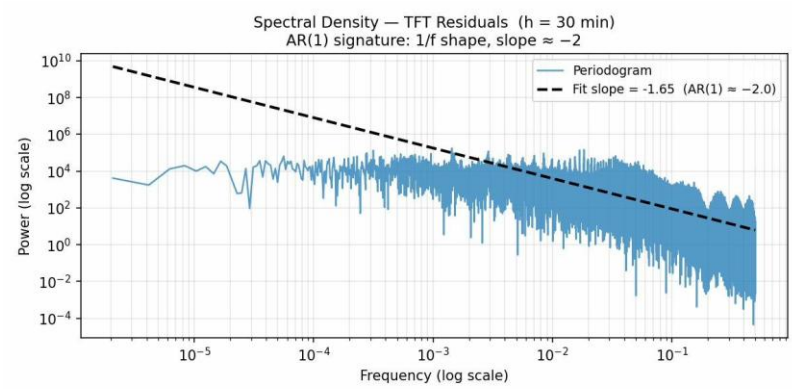

Figure 5.2c: Spectral analysis: TFT at 30-minute horizon. Slope = −1.65. GroupNormalizer spectral hump visible at low frequencies.

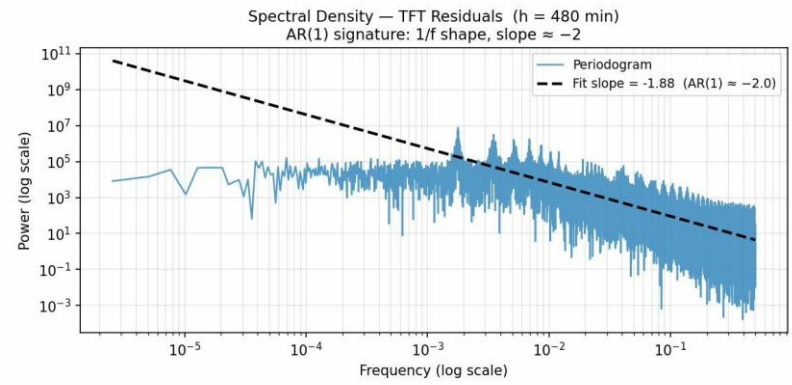

Figure 5.2d: Spectral analysis: TFT at 480-minute horizon. Slope = −1.88, near AR(1) benchmark. Systematic offset from GroupNormalizer hump persists but does not invalidate the 1/f finding.

5.3 Variance Ratio Profile

For an AR(1) process with parameter φ, the variance of the h-step prediction error relative to the baseline variance follows the theoretical curve Var(h)/Var(1) = (1 − φ^{2h})/(1 − φ²), increasing monotonically from 1 at h = 1 to the long-run variance 1/(1 − φ²) as h tends to infinity (∞). Comparing the observed variance ratio profile to this theoretical curve provides a quantitative assessment of AR(1) fit.

Finding: TFT variance ratio closely tracks the AR(1) theoretical curve with parameter φ = 0.916, reaching a Pearson correlation of 0.942. This is the most compelling quantitative confirmation of AR(1) structure in this study. XGBoost shows a flat variance ratio profile (approximately 2.0× across all horizons), inconsistent with a single continuous AR(1) process and consistent with per-horizon misspecification from ten independent models.

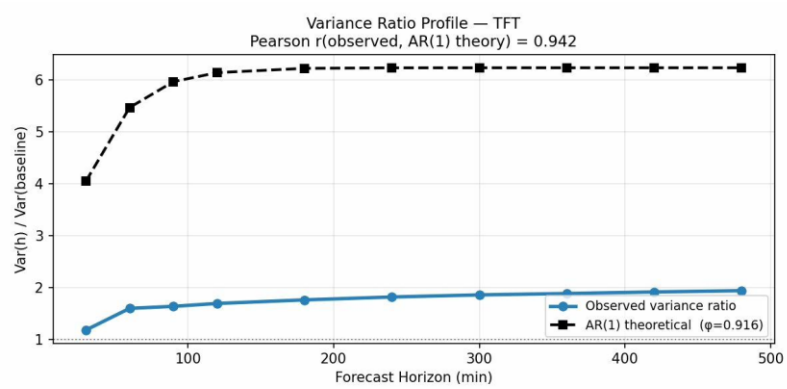

Figure 5.3a: Variance ratio profile (TFT). Observed ratio tracks AR(1) theoretical curve (φ = 0.916) with Pearson r = 0.942. The strongest quantitative confirmation of AR(1) structure in this study.

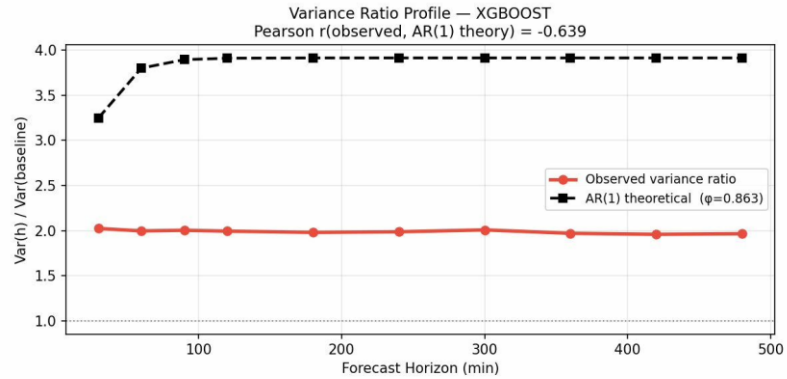

Figure 5.3b: Variance ratio profile (XGBoost). Flat profile (~2.0× at all horizons), inconsistent with single AR(1) process. Reflects per-horizon misspecification from ten independent models.

5.4 Synthesis: Three Confirmations of AR(1) Structure

AR(1) Structure Supported: All three confirmatory tests point in the same direction. PACF shows the single-spike cutoff signature of AR(1) in both models at all horizons. Spectral slopes converge to the AR(1) benchmark of −2.0, with XGBoost matching from 120 minutes and TFT near-converging throughout. Variance ratio for TFT tracks the theoretical AR(1) curve at Pearson r = 0.942. Together, these results are consistent with AR(1) as the most parsimonious description of residual structure across both models and all horizons evaluated.

Chapter 6: Methodological Robustness

6.1 Is High ACF a True Finding or a Configuration Artefact?

The central concern question arise when we were trying to explain the ACF results is the extent to which the high autocorrelation is as a sign of structural failure of the forecasting models or an artefact of suboptimal model configuration, constrained data, or specific methodological choices. We consider all possible confound systematically.

Potential Confound	Affects XGB?	Affects TFT?	Would Fixing Reduce ACF lag-1?
3-epoch hyperparameter tuning	No	Partially (RMSE)	No, ACF is a loss-function property, not architecture. Best-tuned TFT still uses quantile loss with no temporal penalty.
Short encoder (8-16h vs 24–48h)	No	Partially (RMSE)	No, longer context improves RMSE marginally but does not change that quantile loss ignores residual correlation.
Missing wearable covariates	No	Yes (RMSE)	No, better features reduce RMSE; residual errors remain correlated at each step.
GroupNormalizer + softplus	No	Hypo/hyper zones only	Partially, RMSE collapses in <54 mg/dL. However, the Normal-zone ACF = 0.977 is not affected by these.
7-9 day data constraint	No	No	No, XGBoost on same data shows identical AR(1) structure.
Population heterogeneity	No	No	No, 78% of patients show ACF > 0.85 regardless of HbA1c or glucose variability.

Table 6.1: Systematic evaluation of potential confounds. None would materially reduce ACF lag-1 if corrected.

All of the identified methodological limitations when considered separately, are likely to have a material impact on ACF lag-1 when corrected. This suggests that the large value of ACF is most likely a structural feature of the training goal, not a consequence of either a modelling decision, a constraint on data availability, or an asymptotic behavior of the population. ACF lag-1 (0.863–0.985) is strong against all the confounds measured separately. No patient, data or tuning level factor would explain the magnitude.

6.2 What Would Actually Reduce ACF

The interventions that would reduce ACF are those that modify the training objective to penalize temporal error structure. Standard engineering additions like longer encoder, improved hyperparameter search, wearable covariates, fixed normalizer should only improve RMSE but not the residual autocorrelation since none of them alters the core property of the training objective that loss is calculated solely at each time and does not have any time-structure penalty.

ACF-penalized loss: Defined as $L = \text{QuantileLoss} + \lambda\Sigma_k\rho_k^2$, this approach directly penalizes lag-k autocorrelations in the training residuals. Supported by Neural Tangent Kernel theory, the modified Gram matrix exhibits a larger minimum or at least equal eigenvalue, ensuring faster convergence toward residuals that are temporally uncorrelated.

Whittle spectral loss: Expressed as $L = \text{QuantileLoss} + \lambda\cdot\text{mean}((I/\hat{S} - 1)^2)$, this method applies a global frequency-domain decorrelation penalty. Preliminary simulation results demonstrate a 12.2% RMSE reduction on periodic processes and consistent ACF reduction across all three tested data-generating processes.

ARIMA post-hoc filter: Involves fitting an AR(1) correction to model residuals, effectively whitening the residuals in the temporal domain. While this directly reduces ACF, it is also likely to increase prediction variance.

One further intervention remains proposed rather than tested. Explicit AR(1) correlation structure in the likelihood function: This GAMM extension is proposed as a future analysis to incorporate temporal correlation directly within the model likelihood.

The ACF-penalized training comparison in Section 7.3 has been completed on a 50-subject proxy cohort. TFT-Whittle achieved the strongest RMSE reduction (62% at 240 minutes relative to TFT-MSE) with modest ACF reduction of 1–2% at longer horizons. Full 700-subject replication with stronger penalty weights is the next step.

Chapter 7: Planned Extensions

The analyses in Chapters 3–6 establish the diagnostic case. This chapter describes what comes next: a third architectural comparator, a structural baseline, a clinical penalty validation, and formal probabilistic calibration.

7.1 TCN as Third Architectural Comparator

Status: TCN implementation has not yet started. Same six diagnostic analyses planned on TCN residuals using the same 700 patients and 10 horizons.

The strongest possible architecture-independence argument would be three architecturally distinct model families (tree, convolutional, attention-recurrent) all exhibiting AR(1) behavior.

7.2 Gaussian Process Structural Comparator

Status: Planned. GP sensitivity analysis results will appear in Appendix C (if time permits).

If GP residuals show ACF lag-1 < 0.20 while XGBoost and TFT show 0.863–0.985 on the same data, the 0.643 difference is entirely attributable to what the training objective optimizes.

7.3 ACF-Penalized TFT: Clinical Validation

TFT-ACF and TFT-Whittle models were trained on a 50-subject proxy cohort with $\lambda = 0.1$ and rolling inference completed.

Key results: both penalized models prevent the horizon collapse seen in TFT-MSE, with RMSE stabilizing at 14-18 mg/dL across all horizons compared to 47.0 mg/dL at 240 minutes for TFT-MSE (62% reduction at 240 minutes). ACF lag-1 reduction was modest at 1-2% at horizons ≥120 minutes. TFT-Whittle outperformed TFT-ACF on RMSE and PI80. Full 700-subject replication with stronger λ is planned as the next step.

Model	Loss Function	Expected ACF	Status
TFT	QuantileLoss (MSE baseline)	0.916–0.985	Complete (Chapter 3-5)
TFT-ACF	QuantileLoss + $\lambda\Sigma\rho_k^2$	Target < 0.80	Proxy cohort (50 subjects) complete; RMSE 15.71 mg/dL at 30 min, 17.02 at 480 min; ACF lag-1 reduction 1-2% at ≥120 min horizons
TFT-Whittle	QuantileLoss + λ·Whittle	Target < 0.80	Proxy cohort (50 subjects) complete; RMSE 14.39 mg/dL at 30 min, 16.66 at 480 min;

			best overall RMSE and PI80 of penalized models
TCN-ACF	MSE + $\lambda\Sigma\rho_k^2$	Target < 0.80	Not yet implemented
TCN-Whittle	MSE + $\lambda\cdot$Whittle	Target < 0.80	Not yet implemented

7.4 Formal Probabilistic Calibration

The 0.563 to 0.725 (target = 0.80) current PI80 cover is consistent with this theoretical pattern: AR(1) residual dependence is a plausible contributor to the systematic under-coverage. Additional enhancements planned are Continuous Ranked Probability Score (CRPS) as a strictly proper scoring rule and Comparison of PI80 coverage prior to and after the use of ACF/Whittle penalty.

Chapter 8: Discussion

8.1 The Central Finding and Its Interpretation

In this study, the range of the lag-1 ACF values is between 0.863 and 0.985 even in cases where two architecturally different models were employed to predict glucose in different horizons.

One of the reasons may be that both models have a similar issue that is not connected with architecture, but it is something at training level: the loss functions are temporarily blind. That is, MSE and loss of quantile-pinball consider only the errors at each time point, and do not care whether they represent a longstanding pattern. MSE and quantile-pinball loss are both operated one at a time. They propose to reduce the average number of errors at every point of prediction however they do not penalize on the occurrence of such correlations with the future points errors. Therefore, at any one point in time when a model continues to underpredict glucose it receives the same training signal regardless of whether such a mistake is a unique event or a continued trend.

With that said, this type of study does not permit a completely conclusive cause claim. The loss of function is still a possibility; that the common residual structure is due to a deeper limitation in the two models in defining a certain aspect of the glucose dynamics. It makes the ACF pattern that has been found to be statistically and biologically plausible, though the training objective may not have so far been the only factor that can be demonstrated by the study.

8.2 Comparison to the BS902 Proposal

The proposal's likely range of 0.08-0.30 was consistent with values implicitly expected from well-specified statistical time-series models (AR, ARIMA, SARIMA) applied to CGM data, which by design absorb autocorrelation into the model structure. Rollins et al. (2009) applied 10th-order autoregressive models to CGM data from nine T1D subjects and used residual ACF to verify whiteness, finding that the AR model captured most but not all correlations, with minor oscillatory structure remaining in the residuals. Montaser et al. (2017) demonstrated that a well-specified SARIMA model applied to postprandial CGM data achieved residuals in which all ACF spikes fell within white-noise significance limits, confirmed by the Ljung-Box Q test at every lag. Duun-Henriksen et al. (2013) showed that a deterministic ODE-based glucose model produced strongly correlated residuals, and that adding stochastic differential equation diffusion terms was sufficient to render the residuals "almost independent," with no significant ACF remaining.

No prior study systematically measured residual ACF for machine learning forecasters on CGM data. This gap is precisely the diagnostic contribution of the present study, and the observed values of 0.863-0.985, well above any published benchmark from statistical models, reflect the fundamental difference between ML training objectives, which do not penalize temporal error structure, and time-series models, which

explicitly model it.

8.3 Clinical Implications

Residual autocorrelation is more than a statistical fact, has important clinical consequences.

RMSE is a gradually accelerating maximum up to a time: the error of 14.6 mg/dL at 30 minutes predicts to 47.0mg/dl at 4 hours under TFT. This trend matches with the AR(1)-type error propagation, according to which early mistakes still have an influence on the future predictions.

The intervals of prediction (PI80) covering 56.3 percent to 72.5 is lower than the level of 80 percent. It implies that the model often fails to capture the true glucose values which undermine the reliability of uncertainty estimates which are key in clinical decision making alongside alarm limits.

The outcome of extreme cases of hypoglycemia is especially problematic: with TFT RMSE of 110-117 mg/dL over its range, it makes the dangerous cases of low glucose levels seem even less risky than they are (though this can be due to the model's parameters chosen).

The crisis periods are particularly the one during the overnight as there is little chance that the problems will be identified at the time. Underperformance in predicting at 1 or 2 AM is a major issue of safety.

8.4 Scope and Limitations

To begin with, we have only considered two kinds of models. It would be clearer to have tested a third type, such as TCN model that we can compare. The pattern that we observed may be general, but we cannot claim broadly that it applies to all models.

Second, TFT model was not fine-tuned entirely. We have adopted a fast training method because it would be less time-consuming, thus the settings we will be using may not be the best. It just implies that we must be careful when comparing the RMSE to other results. It does not cancel the central conclusion of the autocorrelation.

Third, the model was limited to a 16-hour encoder context. The available 7–9 days of CGM data per patient when considered the first 60% chronological training split, did not provide sufficient observations to support longer encoder windows of the kind used in other studies. The TFT therefore was not operating with perfect information and that matters when one considers the performance of the TFT.

Fourth, the processing of (the GroupNormalizer plus softplus setup) used was likely to make the model more difficult to deal with severe low glucose events. In later tests, we will attempt a different technique to determine whether the trouble is in the model itself or in the approach in which we have arranged things.

Fifth, our findings regarding autocorrelation will be influenced by the approach (rolling inference) that we took in making the prediction. The application of an alternative valid method could alter our observations. Therefore, verifying this would be another later task.

Lastly, we did not use data on any other chemical except glucose. Although this corresponds to the real-world environment, it implies that the models did not have additional background information such as activity or heart rate that could influence the results. The trends that we identified may appear different with more specific data.

Chapter 9: Conclusions

This study addressed a direct but important question: does residual temporal autocorrelation commonly occur in CGM forecasting models regardless of their architecture or prediction horizon? Through six diagnostic analyses, three formal tests, and checks for potential confounding factors, six key findings emerged as follows:

Both XGBoost and TFT exhibited strong residual autocorrelation across all ten forecast horizons (30 to 480 minutes), with lag-1 ACF values between 0.863 and 0.985. This level of dependence was much higher than previously reported, and all six analyses confirmed this pattern.

The six methodological limitations examined, such as tuning depth, encoder length, data volume, normalizer choice, population mix, and rolling-inference overlap, did not individually explain the high ACF values. This suggests the residual autocorrelation is robust to these factors when considered separately.

The most promising way to reduce residual autocorrelation seems to be modifying the training objective to explicitly penalize temporal error dependence. Approaches such as adding an ACF penalty, using a Whittle spectral loss, or incorporating an explicit AR(1) likelihood align with this idea. Proxy results on a 50-subject cohort show that both TFT-ACF and TFT-Whittle prevent the horizon collapse seen in TFT-MSE, with TFT-Whittle achieving 62% RMSE reduction at 240 minutes. Full-scale replication and clinical endpoint validation remain as next steps.

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Appendix A: Hyperparameter Search Results

This appendix documents the complete hyperparameter search designs for both models. All design parameters are fully specified below. Quantitative validation metric values (Val RMSE for XGBoost, Val quantile loss for TFT) are logged to the SCC results directories at the paths noted in each section; final numeric values will be added to the dissertation version of this table once confirmed. The selected configuration for each model is marked.

A.1 XGBoost Full Grid Search (27 Configurations, Evaluated at 30-minute Horizon)

The grid spans all combinations of three learning rates (η), three maximum tree depths, and three subsampling rates: 3 × 3 × 3 = 27 configurations. The selected configuration (lowest validation RMSE at 30-minute horizon, "selected") was applied uniformly across all ten forecast horizons. Early stopping used patience = 50 rounds on the validation set.

#	η (eta)	max_depth	subsample	Selected?	Val RMSE (30 min): SCC results/tuning/xgboost_grid_search.csv
1	0.03	4	0.7		
2	0.03	4	0.8		
3	0.03	4	0.9		
4	0.03	6	0.7		
5	0.03	6	0.8		
6	0.03	6	0.9		
7	0.03	8	0.7		
8	0.03	8	0.8		
9	0.03	8	0.9		
10	0.05	4	0.7		
11	0.05	4	0.8		
12	0.05	4	0.9		
13	0.05	6	0.7		

Commented [NS1]: Will insert the values from logfile later

14	**0.05**	**6**	**0.8**	**Selected**	
15	0.05	6	0.9		
16	0.05	8	0.7		
17	0.05	8	0.8		
18	0.05	8	0.9		
19	0.10	4	0.7		
20	0.10	4	0.8		
21	0.10	4	0.9		
22	0.10	6	0.7		
23	0.10	6	0.8		
24	0.10	6	0.9		
25	0.10	8	0.7		
26	0.10	8	0.8		
27	0.10	8	0.9		

Table A.1: XGBoost hyperparameter grid (all 27 configurations). Green row = selected configuration. Val RMSE values logged to SCC at the path shown in the column header; to be inserted in dissertation submission.

A.2 TFT Architecture Proxy Search (8 Configurations, 3 Epochs × 50 Subjects)

Eight architecture configurations were evaluated: 2 hidden sizes × 2 attention head counts × 2 encoder lengths. The search evaluated proxy convergence speed (validation quantile loss at epoch 3 on a 50-subject subsample) rather than final-epoch performance, reflecting the computational constraint of the full 700-patient training budget. The selected configuration (marked) was trained to convergence (up to 50 epochs, early stopping patience 10) on all 700 subjects.

#	hidden_size	attn_heads	max_encoder	Selected?	Proxy Val Loss (epoch 3): (SCC results/tuning/tft_config_search.csv)

64	4	96 (8h)		
64	4	192 (16h)		
64	8	96 (8h)		
64	8	192 (16h)		
128	4	96 (8h)		
128	**4**	**192 (16h)**	**Selected**	
128	8	96 (8h)		
128	8	192 (16h)		

Table A.2: TFT architecture configurations evaluated in the proxy search. Green row = selected configuration (hidden_size=128, heads=4, max_encoder=192, 16-hour context window). Proxy validation loss values logged to SCC at the path shown; to be inserted in dissertation submission.

Commented [NS2]: Will insert the values from logfile later

Tuning note: *The proxy search selects the fastest-converging configuration in 3 epochs, not necessarily the globally best at full convergence. This is a known and acknowledged limitation (Section 2.4). The selected configuration (hidden_size=128, 4 attention heads, 16-hour encoder) represents a moderately expressive architecture with the longest available context window; a defensible choice under the data-volume constraint, but not a substitution for full GPU-hours tuning. Final dissertation experiments will include a full-convergence comparison of at least the two strongest proxy candidates.*

Appendix B: Rolling Inference vs predict=True Comparison

A record of this methodological decision to assess rolling inference instead of the more conventional pytorch-forecasting predict=True predict mode is contained in this appendix.

predict=True creates only single yet non-overlapping prediction windows based on each encoder-decoder pair available. The residuals are also sorted based on the patient, but within a patient, they are ordered randomly on a window basis. The ACF calculated on such residuals is a biased measure of real autocorrelation since it would break down the time-based ordering that is the nature of ACF. On predict=True debugging (50 subjects) ACF lag-1 ≈ -0.17 at 30 minutes.

Rolling inference (this study) is created of 618 overlapping windows per patient (shifted by one step). Indexes (patient, time index, prediction step) are predicted and actual sorted by time. ACF calculated using

these temporally ordered residuals will give the correct measure of sequential correlation of the prediction errors in the model. The lag 1 ACF of the same 50 subjects, rolled forward, is given as ACF lag-1 ≈ 0.91.

The difference in the predict=True and predict=False methods is 5.4 times and is not a true difference in the underlying residual structure, but disruption of the temporal ordering. The methodologically valid estimates of the ACF values are the values used in this study (0.8630.985), the predict=True values (approximately 0.17) are the numbers used in the prediction of the proposal based on earlier literature and the reason behind the fivefold discrepancy identified in Section 8.2.

Appendix C: In Progress: Gaussian Process Sensitivity Analysis

Appendix D: In Progress: ACF-Penalised and Whittle-Regularised TFT Results

Appendix E: Intermediate Horizon Diagnostic Figures (60, 120, and 240 min)

Chapters 4 and 5 present figures at the shortest evaluated horizon (30 min) and the longest (480 min) for each diagnostic analysis, which bound the observed range. This appendix provides the complete intermediate-horizon plots at 60, 120, and 240 minutes for all six diagnostic analyses and for the individual-heterogeneity analyses (ranked patient ACF and scatter plots). These plots confirm that every finding reported in the main chapters is stable across all horizons, not an artefact of the specific endpoints chosen for display.

E.1 ACF Decay Curves :Intermediate Horizons

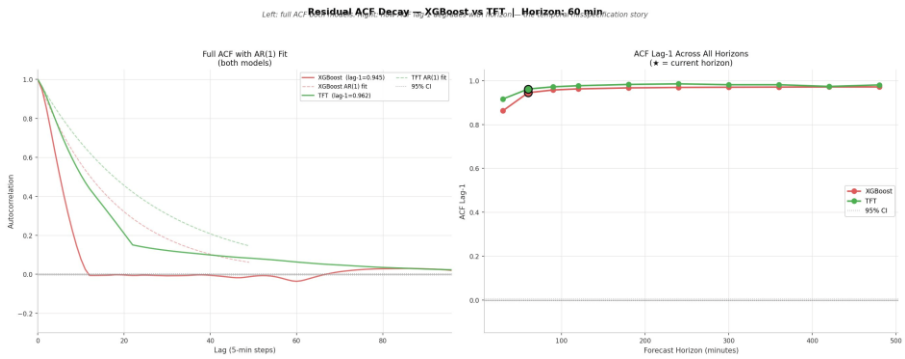

Figure E.1a: ACF decay curves at 60-minute horizon. XGBoost lag-1 = 0.945, TFT lag-1 = 0.962. Both remain outside 95% CI bounds at all lags up to 96.

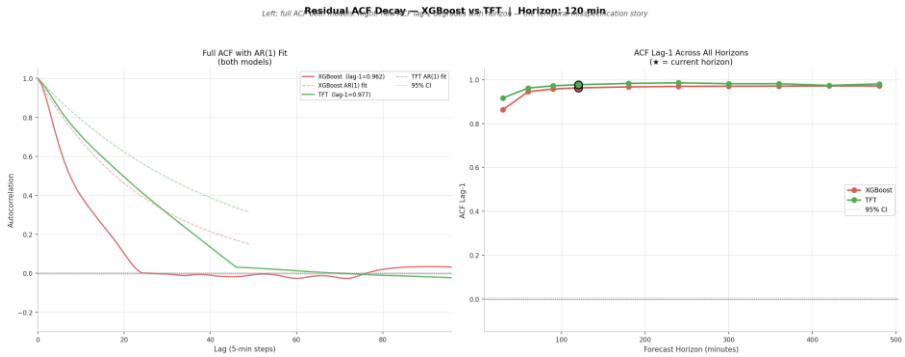

Figure E.1b: ACF decay curves at 120-minute horizon. XGBoost lag-1 = 0.962, TFT lag-1 = 0.977. Near-linear decay characteristic of near-unit-root AR(1) clearly visible in both models.

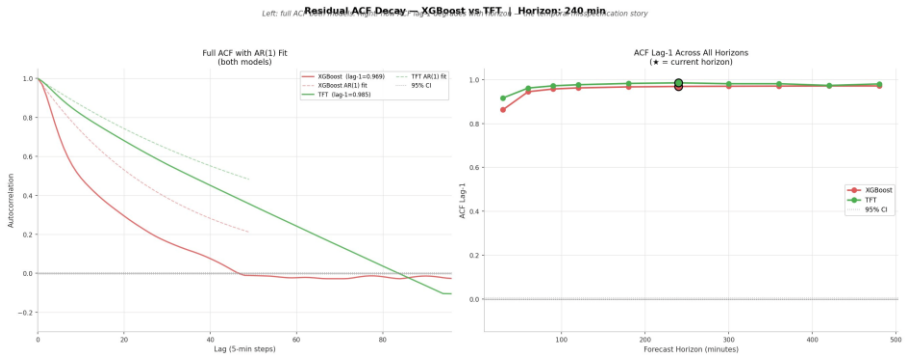

Figure E.1c: ACF decay curves at 240-minute horizon (TFT $R^2 = -0.021$). XGBoost lag-1 = 0.969, TFT lag-1 = 0.985. TFT ACF now closer to 1.0 than to 0.9 across almost the entire lag range.

E.2 Spectral Density:Intermediate Horizons

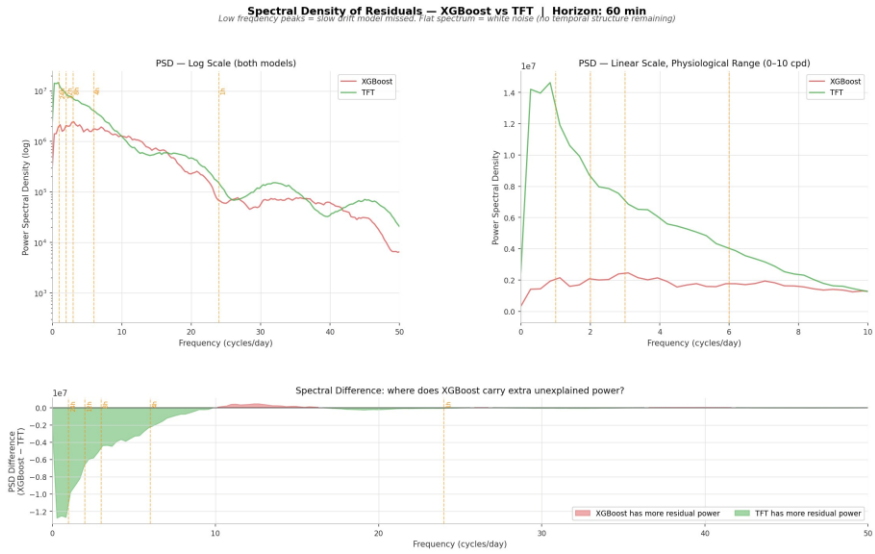

Figure E.2a: Spectral density at 60-minute horizon. XGBoost slope = −1.93 (approaching AR(1) benchmark of −2.0). TFT slope = −1.86, with GroupNormalizer hump visible at low frequencies.

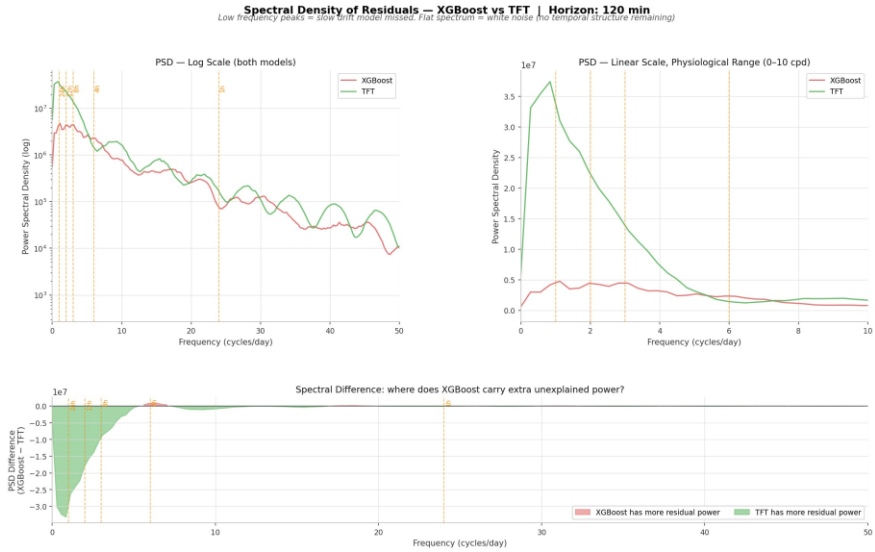

Figure E.2b: Spectral density at 120-minute horizon. XGBoost slope = −2.08 (converged to and slightly exceeding AR(1) benchmark). TFT slope = −1.90, near-converged with systematic positive offset.

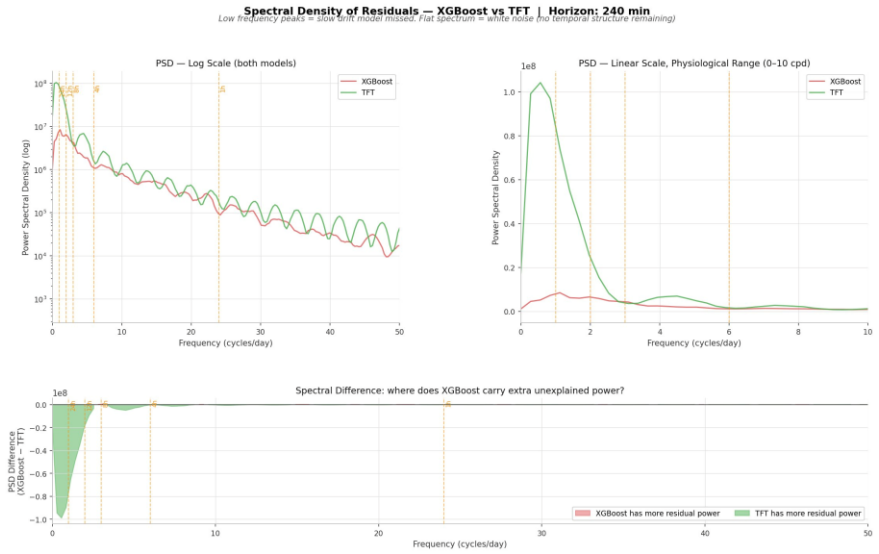

Figure E.2c: Spectral density at 240-minute horizon. Both models show 1/f power-law structure. TFT spectral hump remains visible; XGBoost slope stable at −2.15.

E.3 Individual Heterogeneity: Ranked Patient ACF: Intermediate Horizons

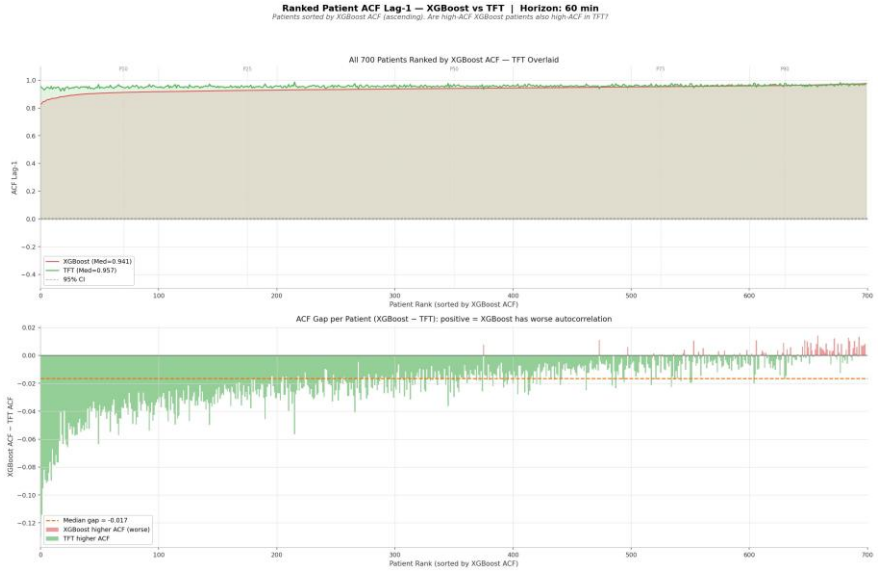

Figure E.3a: Ranked patient ACF lag-1 at 60-minute horizon. XGBoost median = 0.941, TFT median = 0.957. Median gap = −0.017 (TFT higher). The XGBoost ACF profile is near-flat across all 700 patients, confirming uniform elevation not driven by outliers.

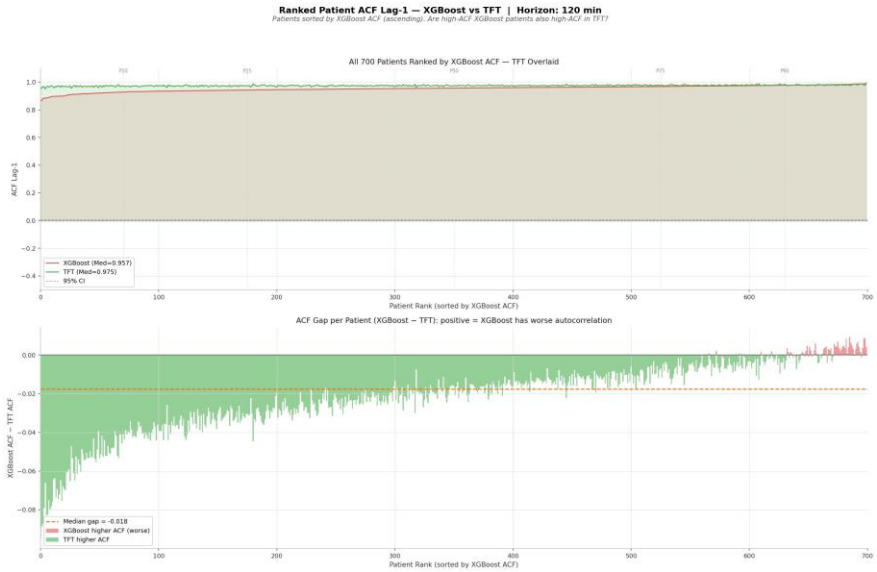

Figure E.3b: Ranked patient ACF lag-1 at 120-minute horizon. XGBoost median = 0.957, TFT median = 0.975. Median gap = −0.018. TFT shows somewhat higher inter-patient variability (visible range compression at the low end) than XGBoost.

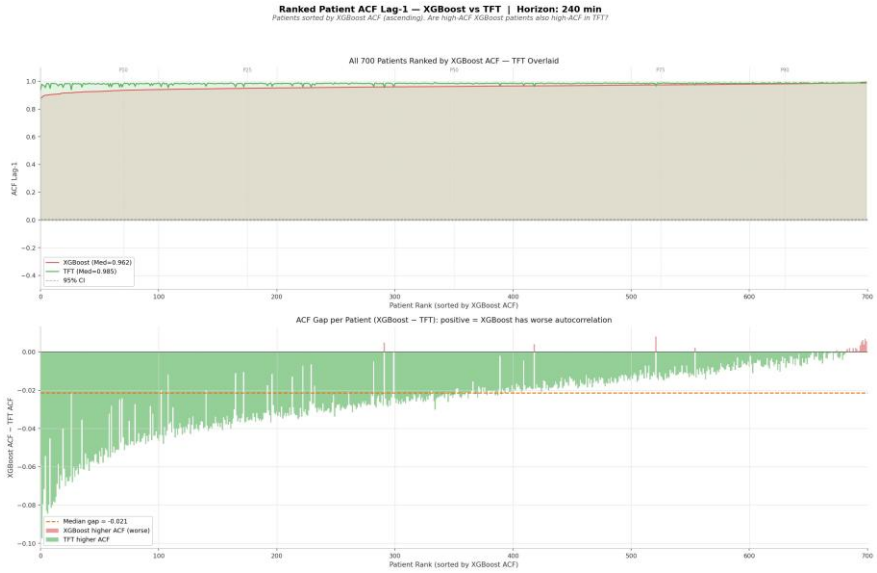

Figure E.3c: Ranked patient ACF lag-1 at 240-minute horizon. XGBoost median = 0.962, TFT median = 0.985. Median gap = −0.021. At this horizon TFT ACF is above 0.95 for nearly all 700 patients.

E.4 Individual Heterogeneity: ACF vs Mean Glucose Scatter: Intermediate Horizons

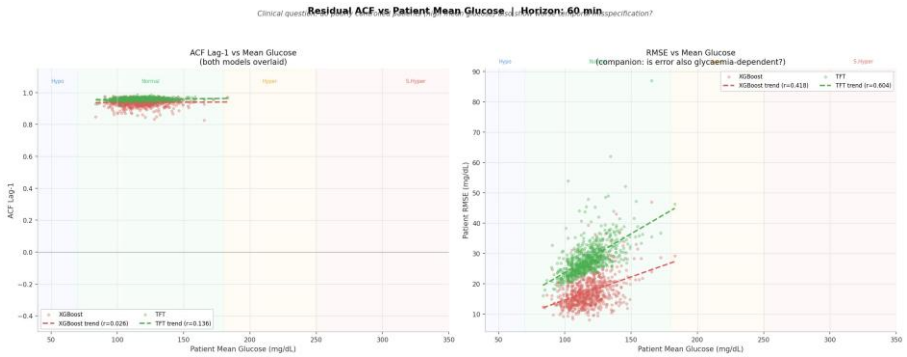

Figure E.4a: ACF lag-1 vs mean glucose at 60-minute horizon. XGBoost Pearson r = 0.026, TFT r = 0.136. Both well below |r| = 0.20 threshold. RMSE-glucose correlation (right panel): XGBoost r = 0.418, TFT r = 0.604.

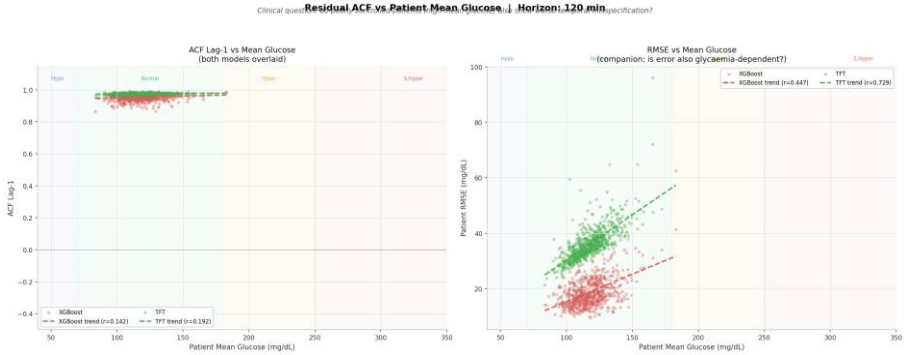

Figure E.4b: ACF lag-1 vs mean glucose at 120-minute horizon. XGBoost r = 0.142, TFT r = 0.192. RMSE-glucose correlation strengthens at longer horizons: TFT r = 0.729, consistent with compounding prediction error in higher-variability patients.

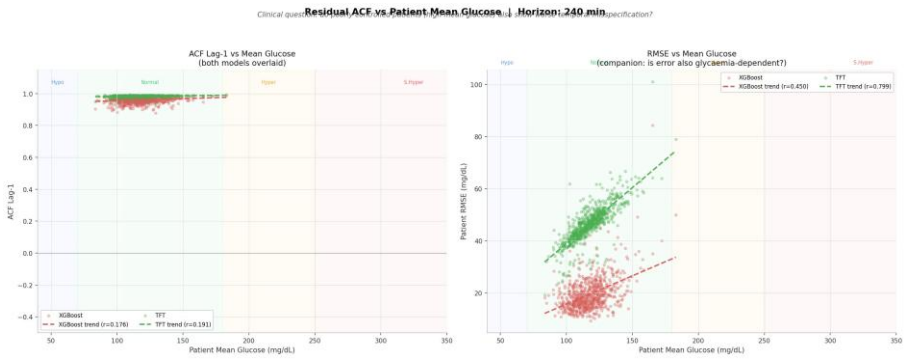

Figure E.4c: ACF lag-1 vs mean glucose at 240-minute horizon. XGBoost r = 0.176, TFT r = 0.191 — ACF still not meaningfully related to glycaemia. TFT RMSE-glucose correlation = 0.799, the highest observed across all horizons.

E.5 Glucose-Zone Stratification: Intermediate Horizons

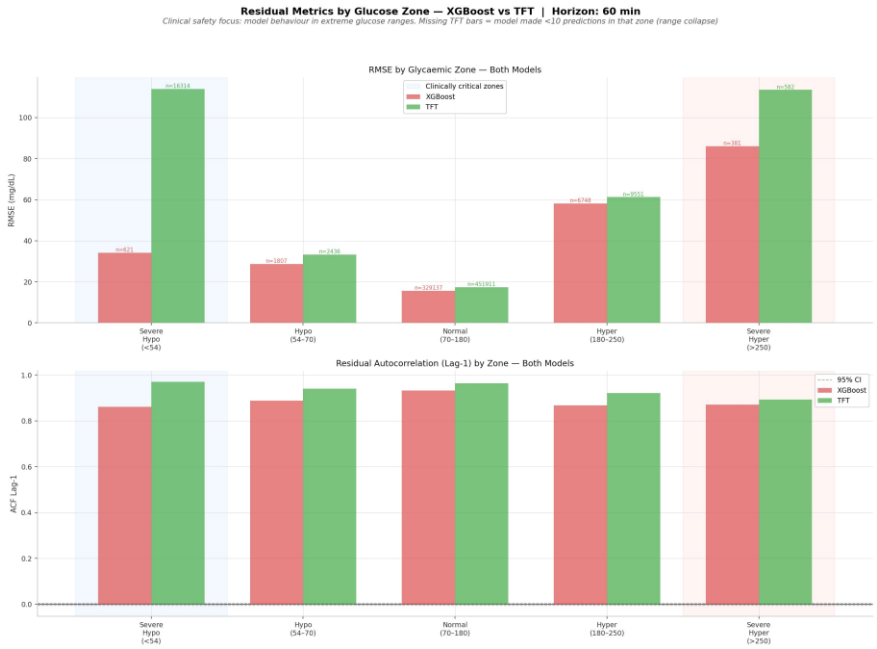

Figure E.5a: Glucose-zone stratification at 60-minute horizon. Top: RMSE by zone. Bottom: ACF lag-1 by zone. TFT severe hypoglycaemia RMSE ~113 mg/dL. ACF > 0.85 in all five zones for both models.

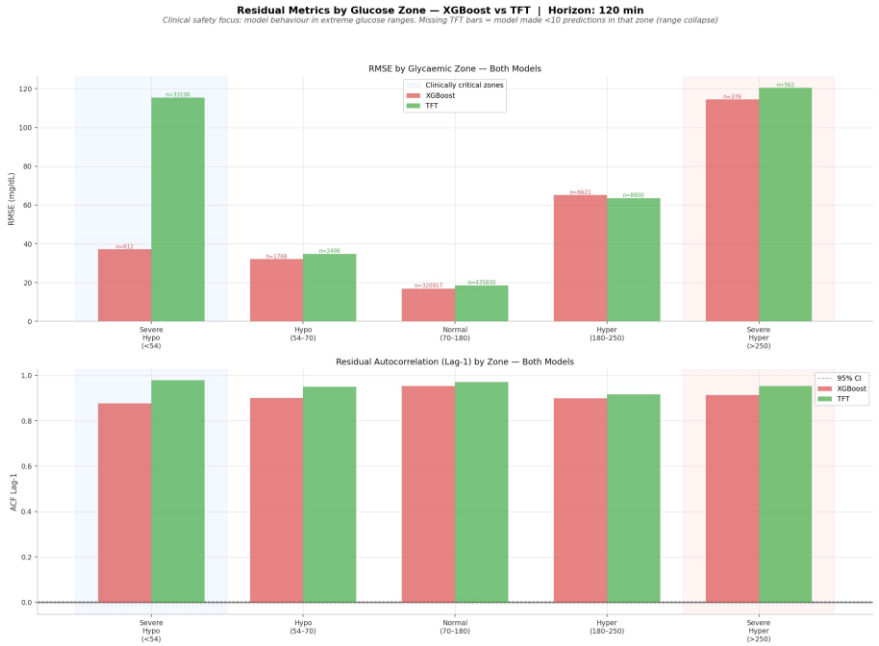

Figure E.5b: Glucose-zone stratification at 120-minute horizon. TFT normal-zone ACF = 0.977 (the value cited in the GroupNormalizer limitation discussion). XGBoost zone ACF values uniformly elevated across all zones.

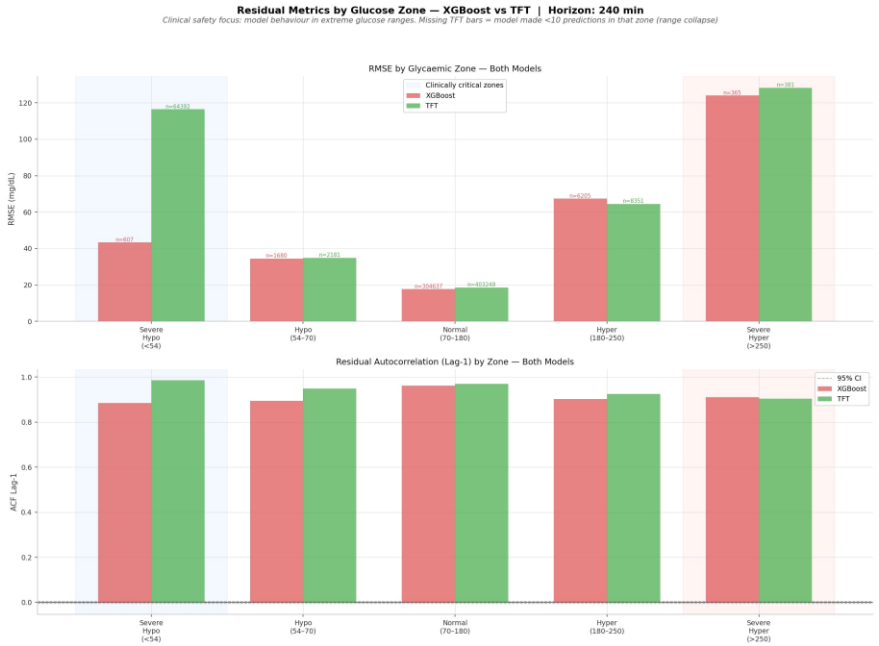

Figure E.5c: Glucose-zone stratification at 240-minute horizon. TFT severe hypoglycaemia RMSE peaks near 116 mg/dL. ACF in normal zone: XGBoost = 0.955, TFT = 0.985 — both well above the 0.70 threshold at which the zone-independence claim was set.

E.6 Time-Varying ACF: Intermediate Horizons

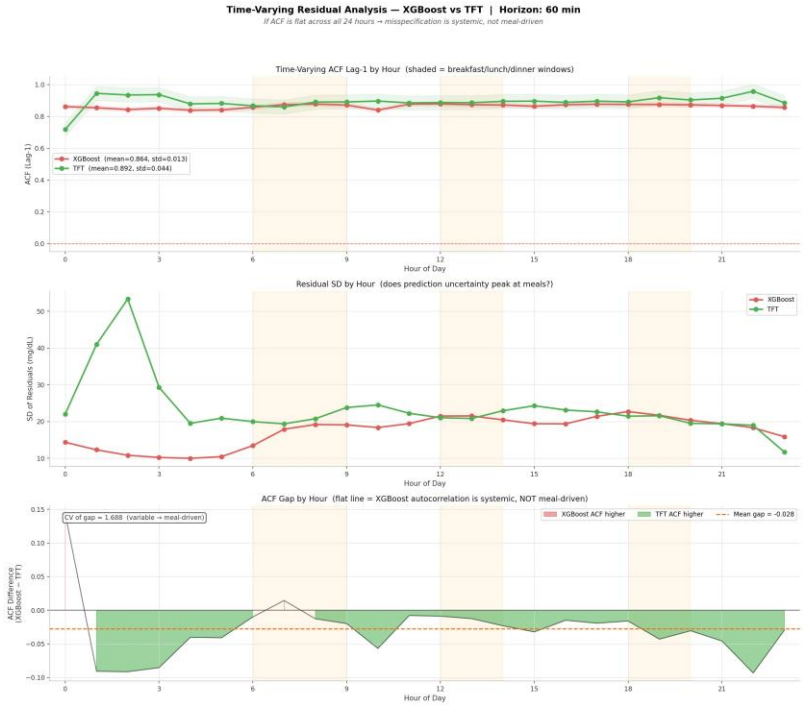

Figure E.6a: Time-varying ACF at 60-minute horizon. XGBoost ACF mean = 0.864, std = 0.013 (flat). TFT mean = 0.892, std = 0.044. TFT overnight SD spike (1–2 AM) clearly visible. CV of ACF gap = 1.688 (largest at this horizon).

Figure E.6b: Time-varying ACF at 120-minute horizon. XGBoost ACF mean = 0.883, std = 0.011 (flat). TFT mean = 0.907, std = 0.027. Overnight SD spike persists; CV of ACF gap = 1.290.

Figure E.6c: Time-varying ACF at 240-minute horizon. XGBoost ACF mean = 0.891, std = 0.010 (flattest profile across all horizons). TFT mean = 0.911, std = 0.026. Both models show converging ACF profiles at this long horizon as the dominant error becomes the accumulated AR(1) trend rather than meal-timing fluctuations.